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Method and apparatus for CSI reporting

Abstract

A method for operating a user equipment (UE) comprises receiving a configuration about a channel state information (CSI) report, the configuration including information about a time interval W for the CSI report, the time interval W comprising multiple time slots; and a number (Y) of channel quality indicator (CQI) values associated with the time interval W; deriving, based on the configuration, the Y CQI values, each of the Y CQI values at a corresponding time slot in the time interval W, wherein the Y CQI values satisfy a block error (BLER) probability requirement; and transmitting the CSI report including the Y CQI values, wherein $Y \geq 1$.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS AND CLAIM OF PRIORITY (1) The present application claims priority to U.S. Provisional Patent Application No. 63/212,373, filed on Jun. 18, 2021, U.S. Provisional Patent Application No. 63/340,829, filed on May 11, 2022, and U.S. Provisional Patent Application No. 63/345,715, filed on May 25, 2022. The content of the above-identified patent document is incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates generally to wireless communication systems and more specifically to CSI reporting.

BACKGROUND

(2) Understanding and correctly estimating the channel between a user equipment (UE) and a base station (BS) (e.g., gNode B (gNB)) is important for efficient and effective wireless communication. In order to correctly estimate the DL channel conditions, the gNB may transmit a reference signal, e.g., CSI-RS, to the UE for DL channel measurement, and the UE may report (e.g., feedback) information about channel measurement, e.g., CSI, to the gNB. With this DL channel measurement, the gNB is able to select appropriate communication parameters to efficiently and effectively perform wireless data communication with the UE.

SUMMARY

(3) Embodiments of the present disclosure provide methods and apparatuses for signaling on CSI format.

(4) In one embodiment, a UE in a wireless communication system is provided. The UE includes a transceiver configured to: receive a configuration about a channel state information (CSI) report, the configuration including information about a time interval W for the CSI report, the time interval W comprising multiple time slots; and a number (Y) of channel quality indicator (CQI) values associated with the time interval W. The UE further includes a processor operably coupled to the transceiver. The processor is configured to derive, based on the configuration, Y CQI values, each at a corresponding time slot in the time interval W, wherein the Y CQI values satisfy a block error (BLER) probability requirement. The transmitter is further configured to transmit the CSI report including the Y CQI values, and $Y \geq 1$.

(5) In another embodiment, a BS in a wireless communication system is provided. The BS includes a processor configured to: generate a configuration about a CSI report, the configuration including information about a time interval W for the CSI report, the time interval W comprising multiple time slots; and a number (Y) of CQI values associated with the time interval W. The BS further includes a transceiver operably coupled to the processor. The transceiver is configured to: transmit the configuration; and receive the CSI report, wherein: the CSI report includes the Y CQI values, the Y CQI values are based on the configuration, each Y CQI value is at a corresponding time slot in the time interval W, the Y CQI values satisfy a BLER probability requirement, and $Y \geq 1$.

(6) In yet another embodiment, a method for operating a UE is provided. The method comprises: receiving a configuration about a CSI report, the configuration including information about a time interval W for the CSI report, the time interval W comprising multiple time slots; and a number (Y) of CQI values associated with the time interval W; deriving, based on the configuration, Y CQI values, each at a corresponding time slot in the time interval W, wherein the Y CQI values satisfy a BLER probability requirement; and transmitting the CSI report including the Y CQI values, wherein $Y \geq 1$.

(7) Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

(8) Before undertaking the DETAILED DESCRIPTION below, it may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The term “couple” and its derivatives refer to any direct or indirect communication between two or more elements, whether or not those elements are in physical contact with one another. The terms “transmit,” “receive,” and “communicate,” as well as derivatives thereof, encompass both direct and indirect communication. The terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation. The term “or” is inclusive, meaning and/or. The phrase “associated with,” as well as derivatives thereof, means to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like. The term “controller” means any device, system or part thereof that controls at least one operation. Such a controller may be implemented in hardware or a combination of hardware and software and/or firmware. The functionality associated with any particular controller may be centralized or distributed, whether locally or remotely. The phrase “at least one of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the following combinations: A, B, C, A and B, A and C, B and C, and A and B and C.

(9) Moreover, various functions described below can be implemented or supported by one or more computer programs, each of which is formed from computer readable program code and embodied in a computer readable medium. The terms “application” and “program” refer to one or more computer programs, software components, sets of instructions, procedures, functions, objects, classes, instances, related data, or a portion thereof adapted for implementation in a suitable computer readable program code. The phrase “computer readable program code” includes any type of computer code, including source code, object code, and executable code. The phrase “computer readable medium” includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. A “non-transitory” computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable memory device.

(10) Definitions for other certain words and phrases are provided throughout this patent document. Those of ordinary skill in the art should understand that in many if not most instances, such definitions apply to prior as well as future uses of such defined words and phrases.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present disclosure and its advantages, reference is now made to the following description taken in conjunction with the accompanying drawings, in which like reference numerals represent like parts:

(2) FIG. 1 illustrates an example wireless network according to embodiments of the present disclosure;

(3) FIG. 2 illustrates an example gNB according to embodiments of the present disclosure;

(4) FIG. 3 illustrates an example UE according to embodiments of the present disclosure;

(5) FIG. 4A illustrates a high-level diagram of an orthogonal frequency division multiple access transmit path according to embodiments of the present disclosure;

(6) FIG. 4B illustrates a high-level diagram of an orthogonal frequency division multiple access receive path according to embodiments of the present disclosure;

(7) FIG. 5 illustrates a transmitter block diagram for a PDSCH in a subframe according to embodiments of the present disclosure;

(8) FIG. 6 illustrates a receiver block diagram for a PDSCH in a subframe according to embodiments of the present disclosure;

(9) FIG. 7 illustrates a transmitter block diagram for a PUSCH in a subframe according to embodiments of the present disclosure;

(10) FIG. 8 illustrates a receiver block diagram for a PUSCH in a subframe according to embodiments of the present disclosure;

(11) FIG. 9 illustrates an example antenna blocks or arrays forming beams according to embodiments of the present disclosure;

(12) FIG. 10 illustrates channel measurements with and without Doppler components according to embodiments of the present disclosure;

(13) FIG. 11 illustrates an antenna port layout according to embodiments of the present disclosure;

(14) FIG. 12 illustrates a 3D grid of oversampled DFT beams according to embodiments of the present disclosure;

(15) FIG. 13 illustrates an example of a UE configured to receive a burst of NZP CSI-RS resources according to embodiments of the present disclosure;

(16) FIG. 14 illustrates an example of a UE configured to determine a value of $N_{\text{sub},4}$ based on the value B in a CSI-RS burst according to embodiments of the present disclosure;

(17) FIG. 15 illustrates an example of a UE configured to determine a value of $N_{\text{sub},4}$ with $J \geq 1$ CSI-RS bursts that occupy a frequency band and a time span according to embodiments of the present disclosure;

(18) FIG. 16 illustrates an example of the time and frequency granularities of reporting X according to embodiments of the present disclosure;

(19) FIG. 17 illustrates an example of the F-T granularity according to embodiments of the present disclosure;

(20) FIG. 18 illustrates a timing relation between CSI-RS occasion, CSI reference resource, and CSI report according to embodiments of the present disclosure;

(21) FIG. 19 illustrates a flow chart of a method for operating a UE according to embodiments of the present disclosure; and

(22) FIG. 20 illustrates a flow chart of a method for operating a BS according to embodiments of the present disclosure.

DETAILED DESCRIPTION

(23) FIG. 1 through FIG. 20, discussed below, and the various embodiments used to describe the principles of the present disclosure in this patent document are by way of illustration only and should not be construed in any way to limit the scope of the disclosure. Those skilled in the art will understand that the principles of the present disclosure may be implemented in any suitably arranged system or device.

(24) The following documents and standards descriptions are hereby incorporated by reference into the present disclosure as if fully set forth herein: 3GPP TS 36.211 v17.0.0, “E-UTRA, Physical channels and modulation” (herein “REF 1”); 3GPP TS 36.212 v17.0.0, “E-UTRA, Multiplexing and Channel coding” (herein “REF 2”); 3GPP TS 36.213 v17.0.0, “E-UTRA, Physical Layer Procedures” (herein “REF 3”); 3GPP TS 36.321 v16.6.0, “E-UTRA, Medium Access Control (MAC) protocol specification” (herein “REF 4”); 3GPP TS 36.331 v16.7.0, “E-UTRA, Radio Resource Control (RRC) protocol specification” (herein “REF 5”); 3GPP TR 22.891 v1.2.0 (herein “REF 6”); 3GPP TS 38.212 v17.0.0, “E-UTRA, NR, Multiplexing and channel coding” (herein “REF 7”); 3GPP TS 38.214 v17.0.0, “E-UTRA, NR, Physical layer procedures for data” (herein “REF 8”); RP-192978, “Measurement results on Doppler spectrum for various UE mobility environments and related CSI enhancements,” Fraunhofer IIS, Fraunhofer HHI, Deutsche Telekom (herein “REF 9”); and 3GPP TS 38.211 v17.0.0, “E-UTRA, NR, Physical channels and modulation (herein “REF 10”).

(25) Aspects, features, and advantages of the disclosure are readily apparent from the following detailed description, simply by illustrating a number

of particular embodiments and implementations, including the best mode contemplated by the disclosure. The disclosure is also capable of other and different embodiments, and its several details can be modified in various obvious respects, all without departing from the spirit and scope of the disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive. The disclosure is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings.

(26) In the following, for brevity, both FDD and TDD are considered as the duplex method for both DL and UL signaling.

(27) Although exemplary descriptions and embodiments to follow assume orthogonal frequency division multiplexing (OFDM) or orthogonal frequency division multiple access (OFDMA), the present disclosure can be extended to other OFDM-based transmission waveforms or multiple access schemes such as filtered OFDM (F-OFDM).

(28) To meet the demand for wireless data traffic having increased since deployment of 4G communication systems, efforts have been made to develop an improved 5G or pre-5G communication system. Therefore, the 5G or pre-5G communication system is also called a “beyond 4G network” or a “post LTE system.”

(29) The 5G communication system is considered to be implemented in higher frequency (mmWave) bands, e.g., 60 GHz bands, so as to accomplish higher data rates or in lower frequency bands, such as below 6 GHz, to enable robust coverage and mobility support. To decrease propagation loss of the radio waves and increase the transmission coverage, the beamforming, massive multiple-input multiple-output (MIMO), full dimensional MIMO (FD-MIMO), array antenna, an analog beam forming, large scale antenna techniques and the like are discussed in 5G communication systems.

(30) In addition, in 5G communication systems, development for system network improvement is under way based on advanced small cells, cloud radio access networks (RANs), ultra-dense networks, device-to-device (D2D) communication, wireless backhaul communication, moving network, cooperative communication, coordinated multi-points (CoMP) transmission and reception, interference mitigation and cancelation and the like.

(31) The discussion of 5G systems and frequency bands associated therewith is for reference as certain embodiments of the present disclosure may be implemented in 5G systems. However, the present disclosure is not limited to 5G systems, or the frequency bands associated therewith, and embodiments of the present disclosure may be utilized in connection with any frequency band. For example, aspects of the present disclosure may also be applied to deployment of 5G communication systems, 6G or even later releases which may use terahertz (THz) bands.

(32) FIGS. 1-4B below describe various embodiments implemented in wireless communications systems and with the use of orthogonal frequency division multiplexing (OFDM) or orthogonal frequency division multiple access (OFDMA) communication techniques. The descriptions of FIGS. 1-3 are not meant to imply physical or architectural limitations to the manner in which different embodiments may be implemented. Different embodiments of the present disclosure may be implemented in any suitably-arranged communications system. The present disclosure covers several components which can be used in conjunction or in combination with one another or can operate as standalone schemes.

(33) FIG. 1 illustrates an example wireless network according to embodiments of the present disclosure. The embodiment of the wireless network shown in FIG. 1 is for illustration only. Other embodiments of the wireless network **100** could be used without departing from the scope of this disclosure.

(34) As shown in FIG. 1, the wireless network includes a gNB **101**, a gNB **102**, and a gNB **103**. The gNB **101** communicates with the gNB **102** and the gNB **103**. The gNB **101** also communicates with at least one network **130**, such as the Internet, a proprietary Internet Protocol (IP) network, or other data network.

(35) The gNB **102** provides wireless broadband access to the network **130** for a first plurality of user equipments (UEs) within a coverage area **120** of the gNB **102**. The first plurality of UEs includes a UE **111**, which may be located in a small business; a UE **112**, which may be located in an enterprise (E); a UE **113**, which may be located in a WiFi hotspot (HS); a UE **114**, which may be located in a first residence (R); a UE **115**, which may be located in a second residence (R); and a UE **116**, which may be a mobile device (M), such as a cell phone, a wireless laptop, a wireless PDA, or the like. The gNB **103** provides wireless broadband access to the network **130** for a second plurality of UEs within a coverage area **125** of the gNB **103**. The second plurality of UEs includes the UE **115** and the UE **116**. In some embodiments, one or more of the gNBs **101-103** may communicate with each other and with the UEs **111-116** using 5G, LTE, LTE-A, WiMAX, WiFi, or other wireless communication techniques.

(36) Depending on the network type, the term “base station” or “BS” can refer to any component (or collection of components) configured to provide wireless access to a network, such as transmit point (TP), transmit-receive point (TRP), an enhanced base station (eNodeB or eNB), a 5G base station (gNB), a macrocell, a femtocell, a WiFi access point (AP), or other wirelessly enabled devices. Base stations may provide wireless access in accordance with one or more wireless communication protocols, e.g., 5G 3GPP new radio interface/access (NR), long term evolution (LTE), LTE advanced (LTE-A), high speed packet access (HSPA), Wi-Fi 802.11a/b/g/n/ac, etc. For the sake of convenience, the terms “BS” and “TRP” are used interchangeably in this patent document to refer to network infrastructure components that provide wireless access to remote terminals. Also, depending on the network type, the term “user equipment” or “UE” can refer to any component such as “mobile station,” “subscriber station,” “remote terminal,” “wireless terminal,” “receive point,” or “user device.” For the sake of convenience, the terms “user equipment” and “UE” are used in this patent document to refer to remote wireless equipment that wirelessly accesses a BS, whether the UE is a mobile device (such as a mobile telephone or smartphone) or is normally considered a stationary device (such as a desktop computer or vending machine).

(37) Dotted lines show the approximate extents of the coverage areas **120** and **125**, which are shown as approximately circular for the purposes of illustration and explanation only. It should be clearly understood that the coverage areas associated with gNBs, such as the coverage areas **120** and **125**, may have other shapes, including irregular shapes, depending upon the configuration of the gNBs and variations in the radio environment associated with natural and man-made obstructions.

(38) As described in more detail below, one or more of the UEs **111-116** include circuitry, programing, or a combination thereof, for receiving a configuration about a channel state information (CSI) report, the configuration including information about a time interval W for the CSI report, the time interval W comprising multiple time slots; and a number (Y) of channel quality indicator (CQI) values associated with the time interval W; deriving, based on the configuration, Y CQI values, each at a corresponding time slot in the time interval W, wherein the Y CQI values satisfy a block error (BLER) probability requirement; and transmitting the CSI report including the Y CQI values, wherein Y≥1. One or more of the gNBs **101-103** includes circuitry, programing, or a combination thereof, for generating a configuration about a channel state information (CSI) report, the configuration including information about a time interval W for the CSI report, the time interval W comprising multiple time slots; and a number (Y) of channel quality indicator (CQI) values associated with the time interval W; and transmitting the configuration; and receiving the CSI report, wherein: the CSI report includes the Y CQI values, the Y CQI values are based on the configuration, each Y CQI value is at a corresponding time slot in the time interval W, the Y CQI values satisfy a block error (BLER) probability requirement, and Y≥1.

(39) Although FIG. 1 illustrates one example of a wireless network, various changes may be made to FIG. 1. For example, the wireless network could include any number of gNBs and any number of UEs in any suitable arrangement. Also, the gNB **101** could communicate directly with any number of UEs and provide those UEs with wireless broadband access to the network **130**. Similarly, each gNB **102-103** could communicate directly with the network **130** and provide UEs with direct wireless broadband access to the network **130**. Further, the gNBs **101**, **102**, and/or **103** could provide access to other or additional external networks, such as external telephone networks or other types of data networks.

(40) FIG. 2 illustrates an example gNB **102** according to embodiments of the present disclosure. The embodiment of the gNB **102** illustrated in FIG. 2 is for illustration only, and the gNBs **101** and **103** of FIG. 1 could have the same or similar configuration. However, gNBs come in a wide variety of configurations, and FIG. 2 does not limit the scope of this disclosure to any particular implementation of a gNB.

(41) As shown in FIG. 2, the gNB **102** includes multiple antennas **205a-205n**, multiple RF transceivers **210a-210n**, transmit (TX) processing circuitry **215**, and receive (RX) processing circuitry **220**. The gNB **102** also includes a controller/processor **225**, a memory **230**, and a backhaul or network interface **235**.

(42) The RF transceivers **210a-210n** receive, from the antennas **205a-205n**, incoming RF signals, such as signals transmitted by UEs in the network **100**. The RF transceivers **210a-210n** down-convert the incoming RF signals to generate IF or baseband signals. The IF or baseband signals are sent to the RX processing circuitry **220**, which generates processed baseband signals by filtering, decoding, and/or digitizing the baseband or IF signals. The RX processing circuitry **220** transmits the processed baseband signals to the controller/processor **225** for further processing.

(43) The TX processing circuitry **215** receives analog or digital data (such as voice data, web data, e-mail, or interactive video game data) from the controller/processor **225**. The TX processing circuitry **215** encodes, multiplexes, and/or digitizes the outgoing baseband data to generate processed baseband or IF signals. The RF transceivers **210a-210n** receive the outgoing processed baseband or IF signals from the TX processing circuitry **215** and up-converts the baseband or IF signals to RF signals that are transmitted via the antennas **205a-205n**.

(44) The controller/processor **225** can include one or more processors or other processing devices that control the overall operation of the gNB **102**. For example, the controller/processor **225** could control the reception of UL channel signals and the transmission of DL channel signals by the RF transceivers **210a-210n**, the RX processing circuitry **220**, and the TX processing circuitry **215** in accordance with well-known principles. The controller/processor **225** could support additional functions as well, such as more advanced wireless communication functions.

(45) For instance, the controller/processor **225** could support beam forming or directional routing operations in which outgoing signals from multiple antennas **205a-205n** are weighted differently to effectively steer the outgoing signals in a desired direction. Any of a wide variety of other functions could be supported in the gNB **102** by the controller/processor **225**.

(46) The controller/processor **225** is also capable of executing programs and other processes resident in the memory **230**, such as an OS. The controller/processor **225** can move data into or out of the memory **230** as required by an executing process.

(47) The controller/processor **225** is also coupled to the backhaul or network interface **235**. The backhaul or network interface **235** allows the gNB **102** to communicate with other devices or systems over a backhaul connection or over a network. The interface **235** could support communications over any suitable wired or wireless connection(s). For example, when the gNB **102** is implemented as part of a cellular communication system (such as one supporting 5G, LTE, or LTE-A), the interface **235** could allow the gNB **102** to communicate with other gNBs over a wired or wireless backhaul connection. When the gNB **102** is implemented as an access point, the interface **235** could allow the gNB **102** to communicate over a wired or wireless local area network or over a wired or wireless connection to a larger network (such as the Internet). The interface **235** includes any suitable structure supporting communications over a wired or wireless connection, such as an Ethernet or RF transceiver.

(48) The memory **230** is coupled to the controller/processor **225**. Part of the memory **230** could include a RAM, and another part of the memory **230** could include a Flash memory or other ROM.

(49) Although FIG. 2 illustrates one example of gNB **102**, various changes may be made to FIG. 2. For example, the gNB **102** could include any number of each component shown in FIG. 2. As a particular example, an access point could include a number of interfaces **235**, and the controller/processor **225** could support routing functions to route data between different network addresses. As another particular example, while shown as including a single instance of TX processing circuitry **215** and a single instance of RX processing circuitry **220**, the gNB **102** could include multiple instances of each (such as one per RF transceiver). Also, various components in FIG. 2 could be combined, further subdivided, or omitted and additional components could be added according to particular needs.

(50) FIG. 3 illustrates an example UE **116** according to embodiments of the present disclosure. The embodiment of the UE **116** illustrated in FIG. 3 is for illustration only, and the UEs **111-115** of FIG. 1 could have the same or similar configuration. However, UEs come in a wide variety of configurations, and FIG. 3 does not limit the scope of this disclosure to any particular implementation of a UE.

(51) As shown in FIG. 3, the UE **116** includes an antenna **305**, a radio frequency (RF) transceiver **310**, TX processing circuitry **315**, a microphone **320**, and receive (RX) processing circuitry **325**. The UE **116** also includes a speaker **330**, a processor **340**, an input/output (I/O) interface (IF) **345**, a touchscreen **350**, a display **355**, and a memory **360**. The memory **360** includes an operating system (OS) **361** and one or more applications **362**.

(52) The RF transceiver **310** receives, from the antenna **305**, an incoming RF signal transmitted by a gNB of the network **100**. The RF transceiver **310** down-converts the incoming RF signal to generate an intermediate frequency (IF) or baseband signal. The IF or baseband signal is sent to the RX processing circuitry **325**, which generates a processed baseband signal by filtering, decoding, and/or digitizing the baseband or IF signal. The RX processing circuitry **325** transmits the processed baseband signal to the speaker **330** (such as for voice data) or to the processor **340** for further processing (such as for web browsing data).

(53) The TX processing circuitry **315** receives analog or digital voice data from the microphone **320** or other outgoing baseband data (such as web data, e-mail, or interactive video game data) from the processor **340**. The TX processing circuitry **315** encodes, multiplexes, and/or digitizes the outgoing baseband data to generate a processed baseband or IF signal. The RF transceiver **310** receives the outgoing processed baseband or IF signal from the TX processing circuitry **315** and up-converts the baseband or IF signal to an RF signal that is transmitted via the antenna **305**.

(54) The processor **340** can include one or more processors or other processing devices and execute the OS **361** stored in the memory **360** in order to control the overall operation of the UE **116**. For example, the processor **340** could control the reception of DL channel signals and the transmission of UL channel signals by the RF transceiver **310**, the RX processing circuitry **325**, and the TX processing circuitry **315** in accordance with well-known principles. In some embodiments, the processor **340** includes at least one microprocessor or microcontroller.

(55) The processor **340** is also capable of executing other processes and programs resident in the memory **360**, such as processes for receiving a configuration about a channel state information (CSI) report, the configuration including information about a time interval W for the CSI report, the time interval W comprising multiple time slots; and a number (Y) of channel quality indicator (CQI) values associated with the time interval W ; deriving, based on the configuration, Y CQI values, each at a corresponding time slot in the time interval W , wherein the Y CQI values satisfy a block error (BLER) probability requirement; and transmitting the CSI report including the Y CQI values, wherein $Y \geq 1$. The processor **340** can move data into or out of the memory **360** as required by an executing process. In some embodiments, the processor **340** is configured to execute the applications **362** based on the OS **361** or in response to signals received from gNBs or an operator. The processor **340** is also coupled to the I/O interface **345**, which provides the UE **116** with the ability to connect to other devices, such as laptop computers and handheld computers. The I/O interface **345** is the communication path between these accessories and the processor **340**.

(56) The processor **340** is also coupled to the touchscreen **350** and the display **355**. The operator of the UE **116** can use the touchscreen **350** to enter data into the UE **116**. The display **355** may be a liquid crystal display, light emitting diode display, or other display capable of rendering text and/or at least limited graphics, such as from web sites.

(57) The memory **360** is coupled to the processor **340**. Part of the memory **360** could include a random-access memory (RAM), and another part of the memory **360** could include a Flash memory or other read-only memory (ROM).

(58) Although FIG. 3 illustrates one example of UE **116**, various changes may be made to FIG. 3. For example, various components in FIG. 3 could be combined, further subdivided, or omitted and additional components could be added according to particular needs. As a particular example, the processor **340** could be divided into multiple processors, such as one or more central processing units (CPUs) and one or more graphics processing units (GPUs). Also, while FIG. 3 illustrates the UE **116** configured as a mobile telephone or smartphone, UEs could be configured to operate as other types of mobile or stationary devices.

(59) FIG. 4A is a high-level diagram of transmit path circuitry. For example, the transmit path circuitry may be used for an orthogonal frequency division multiple access (OFDMA) communication. FIG. 4B is a high-level diagram of receive path circuitry. For example, the receive path circuitry may be used for an orthogonal frequency division multiple access (OFDMA) communication. In FIGS. 4A and 4B, for downlink communication, the transmit path circuitry may be implemented in a base station (gNB) **102** or a relay station, and the receive path circuitry may be implemented in a user equipment (e.g., user equipment **116** of FIG. 1). In other examples, for uplink communication, the receive path circuitry **450** may be

implemented in a base station (e.g., gNB **102** of FIG. 1) or a relay station, and the transmit path circuitry may be implemented in a user equipment (e.g., user equipment **116** of FIG. 1).

(60) Transmit path circuitry comprises channel coding and modulation block **405**, serial-to-parallel (S-to-P) block **410**, Size N Inverse Fast Fourier Transform (IFFT) block **415**, parallel-to-serial (P-to-S) block **420**, add cyclic prefix block **425**, and up-converter (UC) **430**. Receive path circuitry **450** comprises down-converter (DC) **455**, remove cyclic prefix block **460**, serial-to-parallel (S-to-P) block **465**, Size N Fast Fourier Transform (FFT) block **470**, parallel-to-serial (P-to-S) block **475**, and channel decoding and demodulation block **480**.

(61) At least some of the components in FIGS. **4A 400** and **4B 450** may be implemented in software, while other components may be implemented by configurable hardware or a mixture of software and configurable hardware. In particular, it is noted that the FFT blocks and the IFFT blocks described in this disclosure document may be implemented as configurable software algorithms, where the value of Size N may be modified according to the implementation.

(62) Furthermore, although this disclosure is directed to an embodiment that implements the Fast Fourier Transform and the Inverse Fast Fourier Transform, this is by way of illustration only and may not be construed to limit the scope of the disclosure. It may be appreciated that in an alternate embodiment of the present disclosure, the Fast Fourier Transform functions and the Inverse Fast Fourier Transform functions may easily be replaced by discrete Fourier transform (DFT) functions and inverse discrete Fourier transform (IDFT) functions, respectively. It may be appreciated that for DFT and IDFT functions, the value of the N variable may be any integer number (i.e., 1, 4, 3, 4, etc.), while for FFT and IFFT functions, the value of the N variable may be any integer number that is a power of two (i.e., 1, 2, 4, 8, 16, etc.).

(63) In transmit path circuitry **400**, channel coding and modulation block **405** receives a set of information bits, applies coding (e.g., LDPC coding) and modulates (e.g., quadrature phase shift keying (QPSK) or quadrature amplitude modulation (QAM)) the input bits to produce a sequence of frequency-domain modulation symbols. Serial-to-parallel block **410** converts (i.e., de-multiplexes) the serial modulated symbols to parallel data to produce N parallel symbol streams where N is the IFFT/FFT size used in BS **102** and UE **116**. Size N IFFT block **415** then performs an IFFT operation on the N parallel symbol streams to produce time-domain output signals. Parallel-to-serial block **420** converts (i.e., multiplexes) the parallel time-domain output symbols from Size N IFFT block **415** to produce a serial time-domain signal. Add cyclic prefix block **425** then inserts a cyclic prefix to the time-domain signal. Finally, up-converter **430** modulates (i.e., up-converts) the output of add cyclic prefix block **425** to RF frequency for transmission via a wireless channel. The signal may also be filtered at baseband before conversion to RF frequency.

(64) The transmitted RF signal arrives at the UE **116** after passing through the wireless channel, and reverse operations to those at gNB **102** are performed. Down-converter **455** down-converts the received signal to baseband frequency and removes cyclic prefix block **460** and removes the cyclic prefix to produce the serial time-domain baseband signal. Serial-to-parallel block **465** converts the time-domain baseband signal to parallel time-domain signals. Size N FFT block **470** then performs an FFT algorithm to produce N parallel frequency-domain signals. Parallel-to-serial block **475** converts the parallel frequency-domain signals to a sequence of modulated data symbols. Channel decoding and demodulation block **480** demodulates and then decodes the modulated symbols to recover the original input data stream.

(65) Each of gNBs **101-103** may implement a transmit path that is analogous to transmitting in the downlink to user equipment **111-116** and may implement a receive path that is analogous to receiving in the uplink from user equipment **111-116**. Similarly, each one of user equipment **111-116** may implement a transmit path corresponding to the architecture for transmitting in the uplink to gNBs **101-103** and may implement a receive path corresponding to the architecture for receiving in the downlink from gNBs **101-103**.

(66) 5G communication system use cases have been identified and described. Those use cases can be roughly categorized into three different groups. In one example, enhanced mobile broadband (eMBB) is determined to do with high bits/sec requirement, with less stringent latency and reliability requirements. In another example, ultra-reliable and low latency (URLL) is determined with less stringent bits/sec requirement. In yet another example, massive machine type communication (mMTC) is determined that a number of devices can be as many as 100,000 to 1 million per km², but the reliability/throughput/latency requirement could be less stringent. This scenario may also involve power efficiency requirement as well, in that the battery consumption may be minimized as possible.

(67) A communication system includes a downlink (DL) that conveys signals from transmission points such as base stations (BSs) or NodeBs to user equipments (UEs) and an Uplink (UL) that conveys signals from UEs to reception points such as NodeBs. A UE, also commonly referred to as a terminal or a mobile station, may be fixed or mobile and may be a cellular phone, a personal computer device, or an automated device. An eNodeB, which is generally a fixed station, may also be referred to as an access point or other equivalent terminology. For LTE systems, a NodeB is often referred as an eNodeB.

(68) In a communication system, such as LTE system, DL signals can include data signals conveying information content, control signals conveying DL control information (DCI), and reference signals (RS) that are also known as pilot signals. An eNodeB transmits data information through a physical DL shared channel (PDSCH). An eNodeB transmits DCI through a physical DL control channel (PDCCH) or an Enhanced PDCCH (EPDCCH).

(69) An eNodeB transmits acknowledgement information in response to data transport block (TB) transmission from a UE in a physical hybrid ARQ indicator channel (PHICH). An eNodeB transmits one or more of multiple types of RS including a UE-common RS (CRS), a channel state information RS (CSI-RS), or a demodulation RS (DMRS). A CRS is transmitted over a DL system bandwidth (BW) and can be used by UEs to obtain a channel estimate to demodulate data or control information or to perform measurements. To reduce CRS overhead, an eNodeB may transmit a CSI-RS with a smaller density in the time and/or frequency domain than a CRS. DMRS can be transmitted only in the BW of a respective PDSCH or EPDCCH and a UE can use the DMRS to demodulate data or control information in a PDSCH or an EPDCCH, respectively. A transmission time interval for DL channels is referred to as a subframe and can have, for example, duration of 1 millisecond.

(70) DL signals also include transmission of a logical channel that carries system control information. A BCCH is mapped to either a transport channel referred to as a broadcast channel (BCH) when the DL signals convey a master information block (MIB) or to a DL shared channel (DL-SCH) when the DL signals convey a System Information Block (SIB). Most system information is included in different SIBs that are transmitted using DL-SCH. A presence of system information on a DL-SCH in a subframe can be indicated by a transmission of a corresponding PDCCH conveying a codeword with a cyclic redundancy check (CRC) scrambled with system information RNTI (SI-RNTI). Alternatively, scheduling information for a SIB transmission can be provided in an earlier SIB and scheduling information for the first SIB (SIB-1) can be provided by the MIB.

(71) DL resource allocation is performed in a unit of subframe and a group of physical resource blocks (PRBs). A transmission BW includes frequency resource units referred to as resource blocks (RBs). Each RB includes Ng sub-carriers, or resource elements (REs), such as 12 REs. A unit of one RB over one subframe is referred to as a PRB. A UE can be allocated M_{sub}.PDSCH RBs for a total of M_{sub}.sc.sup.PDSCH=M_{sub}.PDSCH.Math.N_{sub}.sc.sup.RB REs for the PDSCH transmission BW.

(72) UL signals can include data signals conveying data information, control signals conveying UL control information (UCI), and UL RS. UL RS includes DMRS and Sounding RS (SRS). A UE transmits DMRS only in a BW of a respective PUSCH or PUCCH. An eNodeB can use a DMRS to demodulate data signals or UCI signals. A UE transmits SRS to provide an eNodeB with an UL CSI. A UE transmits data information or UCI through a respective physical UL shared channel (PUSCH) or a Physical UL control channel (PUCCH). If a UE needs to transmit data information and UCI in a same UL subframe, the UE may multiplex both in a PUSCH. UCI includes Hybrid Automatic Repeat request acknowledgement (HARQ-ACK) information, indicating correct (ACK) or incorrect (NACK) detection for a data TB in a PDSCH or absence of a PDCCH detection (DTX), scheduling request (SR) indicating whether a UE has data in the UE's buffer, rank indicator (RI), and channel state information (CSI) enabling an eNodeB to perform link adaptation for PDSCH transmissions to a UE. HARQ-ACK information is also transmitted by a UE in response to a

detection of a PDCCH/EPDCCH indicating a release of semi-persistent scheduled PDSCH.

(73) An UL subframe includes two slots. Each slot includes $N_{\text{sub.symb.sup.UL}}$ symbols for transmitting data information, UCI, DMRS, or SRS. A frequency resource unit of an UL system BW is an RB. A UE is allocated $N_{\text{sub.RB}}$ RBs for a total of $N_{\text{sub.RB.Math.N.sub.sc.sup.RB}}$ REs for a transmission BW. For a PUCCH, $N_{\text{sub.RB}}=1$. A last subframe symbol can be used to multiplex SRS transmissions from one or more UEs. A number of subframe symbols that are available for data/UCI/DMRS transmission is $N_{\text{sub.symb}}=2 \cdot \text{Math} \cdot (N_{\text{sub.symb.sup.UL}}-1) - N_{\text{sub.SRS}}$, where $N_{\text{sub.SRS}}=1$ if a last subframe symbol is used to transmit SRS and $N_{\text{sub.SRS}}=0$ otherwise.

(74) FIG. 5 illustrates a transmitter block diagram 500 for a PDSCH in a subframe according to embodiments of the present disclosure. The embodiment of the transmitter block diagram 500 illustrated in FIG. 5 is for illustration only. One or more of the components illustrated in FIG. 5 can be implemented in specialized circuitry configured to perform the noted functions or one or more of the components can be implemented by one or more processors executing instructions to perform the noted functions. FIG. 5 does not limit the scope of this disclosure to any particular implementation of the transmitter block diagram 500.

(75) As shown in FIG. 5, information bits 510 are encoded by encoder 520, such as a turbo encoder, and modulated by modulator 530, for example using quadrature phase shift keying (QPSK) modulation. A serial to parallel (S/P) converter 540 generates M modulation symbols that are subsequently provided to a mapper 550 to be mapped to REs selected by a transmission BW selection unit 555 for an assigned PDSCH transmission BW, unit 560 applies an Inverse fast Fourier transform (IFFT), the output is then serialized by a parallel to serial (P/S) converter 570 to create a time domain signal, filtering is applied by filter 580, and a signal transmitted 590. Additional functionalities, such as data scrambling, cyclic prefix insertion, time windowing, interleaving, and others are well known in the art and are not shown for brevity.

(76) FIG. 6 illustrates a receiver block diagram 600 for a PDSCH in a subframe according to embodiments of the present disclosure. The embodiment of the diagram 600 illustrated in FIG. 6 is for illustration only. One or more of the components illustrated in FIG. 6 can be implemented in specialized circuitry configured to perform the noted functions or one or more of the components can be implemented by one or more processors executing instructions to perform the noted functions. FIG. 6 does not limit the scope of this disclosure to any particular implementation of the diagram 600.

(77) As shown in FIG. 6, a received signal 610 is filtered by filter 620, REs 630 for an assigned reception BW are selected by BW selector 635, unit 640 applies a fast Fourier transform (FFT), and an output is serialized by a parallel-to-serial converter 650. Subsequently, a demodulator 660 coherently demodulates data symbols by applying a channel estimate obtained from a DMRS or a CRS (not shown), and a decoder 670, such as a turbo decoder, decodes the demodulated data to provide an estimate of the information data bits 680. Additional functionalities such as time-windowing, cyclic prefix removal, de-scrambling, channel estimation, and de-interleaving are not shown for brevity.

(78) FIG. 7 illustrates a transmitter block diagram 700 for a PUSCH in a subframe according to embodiments of the present disclosure. The embodiment of the block diagram 700 illustrated in FIG. 7 is for illustration only. One or more of the components illustrated in FIG. 5 can be implemented in specialized circuitry configured to perform the noted functions or one or more of the components can be implemented by one or more processors executing instructions to perform the noted functions. FIG. 7 does not limit the scope of this disclosure to any particular implementation of the block diagram 700.

(79) As shown in FIG. 7, information data bits 710 are encoded by encoder 720, such as a turbo encoder, and modulated by modulator 730. A discrete Fourier transform (DFT) unit 740 applies a DFT on the modulated data bits, REs 750 corresponding to an assigned PUSCH transmission BW are selected by transmission BW selection unit 755, unit 760 applies an IFFT and, after a cyclic prefix insertion (not shown), filtering is applied by filter 770 and a signal transmitted 780.

(80) FIG. 8 illustrates a receiver block diagram 800 for a PUSCH in a subframe according to embodiments of the present disclosure. The embodiment of the block diagram 800 illustrated in FIG. 8 is for illustration only. One or more of the components illustrated in FIG. 8 can be implemented in specialized circuitry configured to perform the noted functions or one or more of the components can be implemented by one or more processors executing instructions to perform the noted functions. FIG. 8 does not limit the scope of this disclosure to any particular implementation of the block diagram 800.

(81) As shown in FIG. 8, a received signal 810 is filtered by filter 820. Subsequently, after a cyclic prefix is removed (not shown), unit 830 applies a FFT, REs 840 corresponding to an assigned PUSCH reception BW are selected by a reception BW selector 845, unit 850 applies an inverse DFT (IDFT), a demodulator 860 coherently demodulates data symbols by applying a channel estimate obtained from a DMRS (not shown), a decoder 870, such as a turbo decoder, decodes the demodulated data to provide an estimate of the information data bits 880.

(82) In next generation cellular systems, various use cases are envisioned beyond the capabilities of LTE system. Termed 5G or the fifth-generation cellular system, a system capable of operating at sub-6 GHz and above-6 GHz (for example, in mmWave regime) becomes one of the requirements. In 3GPP TR 22.891, 74 5G use cases have been identified and described; those use cases can be roughly categorized into three different groups. A first group is termed “enhanced mobile broadband (eMBB),” targeted to high data rate services with less stringent latency and reliability requirements. A second group is termed “ultra-reliable and low latency (URLL)” targeted for applications with less stringent data rate requirements, but less tolerant to latency. A third group is termed “massive MTC (mMTC)” targeted for large number of low-power device connections such as 1 million per km^{sup.2} with less stringent the reliability, data rate, and latency requirements.

(83) The 3GPP NR specification supports up to 32 CSI-RS antenna ports which enable a gNB to be equipped with a large number of antenna elements (such as 64 or 128). In this case, a plurality of antenna elements is mapped onto one CSI-RS port. For next generation cellular systems such as 5G, the maximum number of CSI-RS ports can either remain the same or increase.

(84) FIG. 9 illustrates an example antenna blocks or arrays 900 according to embodiments of the present disclosure. The embodiment of the antenna blocks or arrays 1100 illustrated in FIG. 9 is for illustration only. FIG. 9 does not limit the scope of this disclosure to any particular implementation of the antenna blocks or arrays 900.

(85) For mmWave bands, although the number of antenna elements can be larger for a given form factor, the number of CSI-RS ports—which can correspond to the number of digitally precoded ports—tends to be limited due to hardware constraints (such as the feasibility to install a large number of ADCs/DACs at mmWave frequencies) as illustrated in FIG. 9. In this case, one CSI-RS port is mapped onto a large number of antenna elements which can be controlled by a bank of analog phase shifters 901. One CSI-RS port can then correspond to one sub-array which produces a narrow analog beam through analog beamforming 905. This analog beam can be configured to sweep across a wider range of angles (920) by varying the phase shifter bank across symbols or subframes. The number of sub-arrays (equal to the number of RF chains) is the same as the number of CSI-RS ports $N_{\text{sub.CSI-PORT}}$. A digital beamforming unit 910 performs a linear combination across $N_{\text{sub.CSI-PORT}}$ analog beams to further increase precoding gain. While analog beams are wideband (hence not frequency-selective), digital precoding can be varied across frequency sub-bands or resource blocks.

(86) To enable digital precoding, efficient design of CSI-RS is a crucial factor. For this reason, three types of CSI reporting mechanisms corresponding to three types of CSI-RS measurement behavior are supported, for example, “CLASS A” CSI reporting which corresponds to non-precoded CSI-RS, “CLASS B” reporting with $K=1$ CSI-RS resource which corresponds to UE-specific beamformed CSI-RS, and “CLASS B” reporting with $K>1$ CSI-RS resources which corresponds to cell-specific beamformed CSI-RS.

(87) For non-precoded (NP) CSI-RS, a cell-specific one-to-one mapping between CSI-RS port and TXRU is utilized. Different CSI-RS ports have the same wide beam width and direction and hence generally cell wide coverage. For beamformed CSI-RS, beamforming operation, either cell-specific or UE-specific, is applied on a non-zero-power (NZP) CSI-RS resource (e.g., comprising multiple ports). At least at a given time/frequency, CSI-RS ports have narrow beam widths and hence not cell wide coverage, and at least from the gNB perspective. At least some CSI-RS port-resource

combinations have different beam directions.

(88) In scenarios where DL long-term channel statistics can be measured through UL signals at a serving eNodeB, UE-specific BF CSI-RS can be readily used. This is typically feasible when UL-DL duplex distance is sufficiently small. When this condition does not hold, however, some UE feedback is necessary for the eNodeB to obtain an estimate of DL long-term channel statistics (or any of representation thereof). To facilitate such a procedure, a first BF CSI-RS transmitted with periodicity T_1 (ms) and a second NP CSI-RS transmitted with periodicity T_2 (ms), where $T_1 \leq T_2$. This approach is termed hybrid CSI-RS. The implementation of hybrid CSI-RS is largely dependent on the definition of CSI process and NZP CSI-RS resource.

(89) In the 3GPP LTE specification, MIMO has been identified as an essential feature in order to achieve high system throughput requirements and it will continue to be the same in NR. One of the key components of a MIMO transmission scheme is the accurate CSI acquisition at the eNB (or TRP). For MU-MIMO, in particular, the availability of accurate CSI is necessary in order to guarantee high MU performance. For TDD systems, the CSI can be acquired using the SRS transmission relying on the channel reciprocity. For FDD systems, on the other hand, the CSI can be acquired using the CSI-RS transmission from the eNB, and CSI acquisition and feedback from the UE. In legacy FDD systems, the CSI feedback framework is ‘implicit’ in the form of CQI/PMI/RI derived from a codebook assuming SU transmission from the eNB. Because of the inherent SU assumption while deriving CSI, this implicit CSI feedback is inadequate for MU transmission. Since future (e.g., NR) systems are likely to be more MU-centric, this SU-MU CSI mismatch will be a bottleneck in achieving high MU performance gains. Another issue with implicit feedback is the scalability with larger number of antenna ports at the eNB. For large number of antenna ports, the codebook design for implicit feedback is quite complicated, and the designed codebook is not guaranteed to bring justifiable performance benefits in practical deployment scenarios (for example, only a small percentage gain can be shown at the most).

(90) In 5G or NR systems, the above-mentioned CSI reporting paradigm from LTE is also supported and referred to as Type I CSI reporting. In addition to Type I, a high-resolution CSI reporting, referred to as Type II CSI reporting, is also supported to provide more accurate CSI information to gNB for use cases such as high-order MU-MIMO. The overhead of Type II CSI reporting can be an issue in practical UE implementations. One approach to reduce Type II CSI overhead is based on frequency domain (FD) compression. In Rel. 16 NR, DFT-based FD compression of the Type II CSI has been supported (referred to as Rel. 16 enhanced Type II codebook in REF8). Some of the key components for this feature includes (a) spatial domain (SD) basis $W_{sub.1}$, (b) FD basis $W_{sub.f}$, and (c) coefficients $\{\tilde{w}\}_{sub.2}$ that linearly combine SD and FD basis. In a non-reciprocal FDD system, a complete CSI (comprising all components) needs to be reported by the UE. However, when reciprocity or partial reciprocity does exist between UL and DL, then some of the CSI components can be obtained based on the UL channel estimated using SRS transmission from the UE. In Rel. 16 NR, the DFT-based FD compression is extended to this partial reciprocity case (referred to as Rel. 16 enhanced Type II port selection codebook in REF8), wherein the DFT-based SD basis in $W_{sub.1}$ is replaced with SD CSI-RS port selection, i.e., L out of $\frac{P_{CSI-RS}}{2}$

CSI-RS ports are selected (the selection is common for the two antenna polarizations or two halves of the CSI-RS ports). The CSI-RS ports in this case are beamformed in SD (assuming UL-DL channel reciprocity in angular domain), and the beamforming information can be obtained at the gNB based on UL channel estimated using SRS measurements.

(92) It has been known in the literature that UL-DL channel reciprocity exists in both angular and delay domains if the UL-DL duplexing distance is small. Since delay in time domain transforms (or closely related to) basis vectors in frequency domain (FD), the Rel. 16 enhanced Type II port selection can be further extended to both angular and delay domains (or SD and FD). In particular, the DFT-based SD basis in $W_{sub.1}$ and/or DFT-based FD basis in $W_{sub.f}$ can be replaced with SD and FD port selection, i.e., L CSI-RS ports are selected in SD and/or M ports are selected in FD. The CSI-RS ports in this case are beamformed in SD (assuming UL-DL channel reciprocity in angular domain) and/or FD (assuming UL-DL channel reciprocity in delay/frequency domain), and the corresponding SD and/or FD beamforming information can be obtained at the gNB based on UL channel estimated using SRS measurements. In Rel. 17 NR, such a codebook will be supported.

(93) FIG. 10 illustrates channel measurement with and without Doppler components 1000 according to embodiments of the present disclosure. The embodiment of the channel measurement with and without Doppler components 1000 illustrated in FIG. 10 is for illustration only. FIG. 10 does not limit the scope of this disclosure to any particular implementation of the channel measurement with and without Doppler components 1000.

(94) Now, when the UE speed is in a moderate or high-speed regime, the performance of the Rel. 15/16/17 codebooks starts to deteriorate quickly due to fast channel variations (which in turn is due to UE mobility that contributes to the Doppler component of the channel), and a one-shot nature of CSI-RS measurement and CSI reporting in Rel. 15/16/17. This limits the usefulness of Rel. 15/16/17 codebooks to low mobility or static UEs only. For moderate or high mobility scenarios, an enhancement in CSI-RS measurement and CSI reporting is needed, which is based on the Doppler components of the channel. As described in [REF9], the Doppler components of the channel remain almost constant over a large time duration, referred to as channel stationarity time, which is significantly larger than the channel coherence time. Note that the current (Rel. 15/16/17) CSI reporting is based on the channel coherence time, which is not suitable when the channel has significant Doppler components. The Doppler components of the channel can be calculated based on measuring a reference signal (RS) burst, where the RS can be CSI-RS or SRS. When the RS is CSI-RS, the UE measures a CSI-RS burst, and use it to obtain Doppler components of the DL channel, and when RS is SRS, the gNB measures an SRS burst, and use it to obtain Doppler components of the UL channel. The obtained Doppler components can be reported by the UE using a codebook (as part of a CS report). Or the gNB can use the obtained Doppler components of the UL channel to beamform CSI-RS for CSI reporting by the UE. An illustration of channel measurement with and without Doppler components is shown in FIG. 10. When the channel is measured with the Doppler components (e.g., based on an RS burst), the measured channel can remain close to the actual varying channel. On the other hand, when the channel is measured without the Doppler components (e.g., based on a one-shot RS), the measured channel can be far from the actual varying channel.

(95) As described, measuring an RS burst is needed in order to obtain the Doppler components of the channel. This disclosure provides several example embodiments on obtaining the Doppler domain components or units that determine the length of the basis vectors that are used for the Doppler compression. The disclosure also describes example embodiments on signaling related to the CSI reporting format.

(96) All the following components and embodiments are applicable for UL transmission with CP-OFDM (cyclic prefix OFDM) waveform as well as DFT-SOFDM (DFT-spread OFDM) and SC-FDMA (single-carrier FDMA) waveforms. Furthermore, all the following components and embodiments are applicable for UL transmission when the scheduling unit in time is either one subframe (which can consist of one or multiple slots) or one slot.

(97) In the present disclosure, the frequency resolution (reporting granularity) and span (reporting bandwidth) of CSI reporting can be defined in terms of frequency “subbands” and “CSI reporting band” (CRB), respectively.

(98) A subband for CSI reporting is defined as a set of contiguous PRBs which represents the smallest frequency unit for CSI reporting. The number of PRBs in a subband can be fixed for a given value of DL system bandwidth, configured either semi-statically via higher-layer/RRC signaling, or dynamically via L1 DL control signaling or MAC control element (MAC CE). The number of PRBs in a subband can be included in CSI reporting setting.

(99) “CSI reporting band” is defined as a set/collection of subbands, either contiguous or non-contiguous, wherein CSI reporting is performed. For example, CSI reporting band can include all the subbands within the DL system bandwidth. This can also be termed “full-band”. Alternatively, CSI reporting band can include only a collection of subbands within the DL system bandwidth. This can also be termed “partial band”.

(100) The term “CSI reporting band” is used only as an example for representing a function. Other terms such as “CSI reporting subband set” or “CSI reporting bandwidth” can also be used.

(101) In terms of UE configuration, a UE can be configured with at least one CSI reporting band. This configuration can be semi-static (via higher-

layer signaling or RRC) or dynamic (via MAC CE or L1 DL control signaling). When configured with multiple (N) CSI reporting bands (e.g., via RRC signaling), a UE can report CSI associated with $n \leq N$ CSI reporting bands. For instance, >6 GHz, large system bandwidth may require multiple CSI reporting bands. The value of n can either be configured semi-statically (via higher-layer signaling or RRC) or dynamically (via MAC CE or L1 DL control signaling). Alternatively, the UE can report a recommended value of n via an UL channel.

(102) Therefore, CSI parameter frequency granularity can be defined per CSI reporting band as follows. A CSI parameter is configured with “single” reporting for the CSI reporting band with M.sub.n subbands when one CSI parameter for all the M.sub.n subbands within the CSI reporting band. A CSI parameter is configured with “subband” for the CSI reporting band with M.sub.n subbands when one CSI parameter is reported for each of the M.sub.n subbands within the CSI reporting band.

(103) FIG. 11 illustrates an example antenna port layout 1100 according to embodiments of the present disclosure. The embodiment of the antenna port layout 1100 illustrated in FIG. 11 is for illustration only. FIG. 11 does not limit the scope of this disclosure to any particular implementation of the antenna port layout 1100.

(104) As illustrated in FIG. 11, N.sub.1 and N.sub.2 are the number of antenna ports with the same polarization in the first and second dimensions, respectively. For 2D antenna port layouts, N.sub.1>1, N.sub.2>1, and for 1D antenna port layouts N.sub.1>1 and N.sub.2=1. Therefore, for a dual-polarized antenna port layout, the total number of antenna ports is 2N.sub.1N.sub.2.

(105) As described in U.S. Pat. No. 10,659,118, issued May 19, 2020, and entitled “Method and Apparatus for Explicit CSI Reporting in Advanced Wireless Communication Systems,” which is incorporated herein by reference in its entirety, a UE is configured with high-resolution (e.g., Type II) CSI reporting in which the linear combination-based Type II CSI reporting framework is extended to include a frequency dimension in addition to the first and second antenna port dimensions.

(106) FIG. 12 illustrates a 3D grid 1300 of the oversampled DFT beams (1st port dim, 2nd port dim., freq. dim.) in which 1st dimension is associated with the 1st port dimension, 2nd dimension is associated with the 2nd port dimension, and 3rd dimension is associated with the frequency dimension.

(107) The basis sets for 1.sup.st and 2.sup.nd port domain representation are oversampled DFT codebooks of length-N.sub.1 and length-N.sub.2, respectively, and with oversampling factors O.sub.1 and O.sub.2, respectively. Likewise, the basis set for frequency domain representation (i.e., 3rd dimension) is an oversampled DFT codebook of length-N.sub.3 and with oversampling factor O.sub.3. In one example, O.sub.1=O.sub.2=O.sub.3=4. In another example, the oversampling factors O.sub.i belongs to {2, 4, 8}. In yet another example, at least one of O.sub.1, O.sub.2, and O.sub.3 is higher layer configured (via RRC signaling).

(108) As explained in Section 5.2.2.2.6 of REFS, a UE is configured with higher layer parameter codebookType set to ‘typeII-PortSelection-r16’ for an enhanced Type II CSI reporting in which the pre-coders for all SBs and for a given layer l=1, . . . , v, where v is the associated RI value, is given by either

(109)

$$W^l = A C_l B^H = \begin{bmatrix} a_0 & a_1 & \dots & a_{L-1} \end{bmatrix} \begin{bmatrix} c_{l,0,0} & c_{l,0,1} & \dots & c_{l,0,M-1} \\ c_{l,1,0} & c_{l,1,1} & \dots & c_{l,1,M-1} \\ \dots & \dots & \dots & \dots \\ c_{l,L-1,0} & c_{l,L-1,1} & \dots & c_{l,L-1,M-1} \end{bmatrix} \begin{bmatrix} b_0 & b_1 & \dots & b_{M-1} \end{bmatrix} = \sum_{f=0}^{M-1} \sum_{i=0}^{L-1} c_{l,i,f} (a_i b_f^H)$$

or

$$W^l = \begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix} C_l B^H = \begin{bmatrix} a_0 & a_1 & \dots & a_{L-1} & 0 \\ 0 & a_0 & a_1 & \dots & a_{L-1} \end{bmatrix} \begin{bmatrix} c_{l,0,0} & c_{l,0,1} & \dots & c_{l,0,M-1} \\ c_{l,1,0} & c_{l,1,1} & \dots & c_{l,1,M-1} \\ \dots & \dots & \dots & \dots \\ c_{l,L-1,0} & c_{l,L-1,1} & \dots & c_{l,L-1,M-1} \end{bmatrix} \begin{bmatrix} b_0 & b_1 & \dots & b_{M-1} \end{bmatrix}^H$$

where N.sub.1 is a number of antenna ports in a first antenna port dimension (having the same antenna polarization), N.sub.2 is a number of antenna ports in a second antenna port dimension (having the same antenna polarization), P.sub.CSI-RS is a number of CSI-RS ports configured to the UE, N.sub.3 is a number of SBs for PMI reporting or number of FD units or number of FD components (that comprise the CSI reporting band) or a total number of precoding matrices indicated by the PMI (one for each FD unit/component), a.sub.i is a 2N.sub.1N.sub.2×1 (Eq. 1) or N.sub.1N.sub.2×1 (Eq. 2) column vector, and a.sub.i is a N.sub.1N.sub.2×1 or

(110) $\frac{P_{\text{CSI-RS}}}{2} \times 1$

port selection column vector if antenna ports at the gNB are co-polarized, and is a 2N.sub.1N.sub.2×1 or P.sub.CSIRS×1 port selection column vector if antenna ports at the gNB are dual-polarized or cross-polarized, where a port selection vector is defined as a vector which contains a value of 1 in one element and zeros elsewhere, and P.sub.CSIRS is the number of CSI-RS ports configured for CSI reporting, b.sub.f is a N.sub.3×1 column vector, c.sub.l,i,f is a complex coefficient associate with vectors a.sub.i and b.sub.f.

(111) In a variation, when the UE reports a subset K<2LM coefficients (where K is either fixed, configured by the gNB or reported by the UE), then the coefficient c.sub.l,i,f in precoder equations Eq. 1 or Eq. 2 is replaced with x.sub.l,i,f×c.sub.l,i,f, where x.sub.l,i,f=1 if the coefficient c.sub.l,i,f is reported by the UE according to some embodiments of this disclosure. x.sub.l,i,f=0 otherwise (i.e., c.sub.l,i,f is not reported by the UE).

(112) The indication whether x.sub.l,i,f=1 or 0 is according to some embodiments of this disclosure. For example, it can be via a bitmap.

(113) In a variation, the precoder equations Eq. 1 or Eq. 2 are respectively generalized to

$$(114) W^l = \sum_{f=0}^{M-1} \sum_{i=0}^{L-1} c_{l,i,f} (a_i b_f^H) \quad (\text{Eq. 3})$$

and

$$(115) W^l = \begin{bmatrix} \sum_{i=0}^{L-1} \sum_{f=0}^{M-1} c_{l,i,f} (a_i b_f^H) \\ \sum_{i=0}^{L-1} \sum_{f=0}^{M-1} c_{l,i+L,f} (a_i b_f^H) \end{bmatrix}, \quad (\text{Eq. 4})$$

where for a given i, the number of basis vectors is M.sub.i and the corresponding basis vectors are {b.sub.i,f}. Note that M.sub.i is the number of coefficients c.sub.l,i,f reported by the UE for a given i, where M.sub.i≤M (where {M.sub.i} or ΣM.sub.i is either fixed, configured by the gNB or reported by the UE).

(116) The columns of W.sup.l are normalized to norm one. For rank R or R layers (v=R), the pre-coding matrix is given by

$$(117) W^{(R)} = \frac{1}{\sqrt{R}} \begin{bmatrix} W^1 & W^2 & \dots & W^R \end{bmatrix}.$$

Eq. 2 is assumed in the rest of the disclosure. The embodiments of the disclosure, however, are general and are also application to Eq. 1, Eq. 3, and Eq. 4.

(118) Here

$$(119) L \leq \frac{P_{\text{CSI-RS}}}{2} \text{ and } M \leq N_3. \text{ If } L = \frac{P_{\text{CSI-RS}}}{2},$$

then A is an identity matrix, and hence not reported. Likewise, if M=N.sub.3, then B is an identity matrix, and hence not reported. Assuming M<N.sub.3, in an example, to report columns of B, the oversampled DFT codebook is used. For instance, b.sub.f=w.sub.f, where the quantity w.sub.f is given by

$$(120) \mathbf{w}_f = \begin{bmatrix} 1 & e^{j^2 \frac{n_3^0}{O_3 N_3}} & e^{j^2 \frac{\text{Math. } 2n_3^0}{O_3 N_3}} & \dots & e^{j^2 \frac{\text{Math. } (N_3 - 1)n_3^0}{O_3 N_3}} \end{bmatrix}^T.$$

(121) When $O_{\text{sub}.3}=1$, the FD basis vector for layer $l \in \{1, \dots, v\}$ (where v is the RI or rank value) is given by $\mathbf{w}_{\text{sub}.f} = [y_{\text{sub}.0,l,\text{sup}.(f)} y_{\text{sub}.1,l,\text{sup}.(f)} \dots y_{\text{sub}.N_{\text{sub}.3,\text{sub}.1,\text{sup}.(f)},\text{sup}.T}]$, where

$$(122) y_{l,l}^{(f)} = e^{j^2 \frac{n_3^0}{N_3}}$$

and $n_{\text{sub}.3,l} = [n_{\text{sub}.3,l,\text{sup}.(0)}, \dots, n_{\text{sub}.3,l,\text{sup}.(M-1)}]$ where $n_{\text{sub}.3,l,\text{sup}.(f)} \in \{0, 1, \dots, N_{\text{sub}.3}-1\}$.

(124) In another example, discrete cosine transform DCT basis is used to construct/report basis B for the 3^{sup}.rd dimension. The m-th column of the DCT compression matrix is simply given by

$$(125) 0[W_f]_{nm} = \begin{cases} \frac{1}{\sqrt{K}}, & n = 0 \\ \sqrt{\frac{2}{K}} \cos\left(\frac{(2m+1)n}{2K}\pi\right), & n = 1, \dots, \text{Math. } K - 1 \end{cases}, \text{ and } K = N_3, \text{ and } m = 0, \dots, \text{Math. } N_3 - 1.$$

(126) Since DCT is applied to real valued coefficients, the DCT is applied to the real and imaginary components (of the channel or channel eigenvectors) separately. Alternatively, the DCT is applied to the magnitude and phase components (of the channel or channel eigenvectors) separately. The use of DFT or DCT basis is for illustration purpose only. The disclosure is applicable to any other basis vectors to construct/report A and B.

(127) On a high level, a precoder $\mathbf{W}_{\text{sup}.l}$ can be described as follows.

$$\mathbf{W} = \mathbf{A} \cdot \mathbf{W}_{\text{sub}.l} \cdot \mathbf{B}, \text{sub}.l,\text{sup}.H = \mathbf{W}_{\text{sub}.1} \{\tilde{\text{over}}(W)\}_{\text{sub}.2} \mathbf{W}_{\text{sub}.f,\text{sup}.H}, \quad (\text{Eq. 5})$$

where $\mathbf{A} = \mathbf{W}_{\text{sub}.1}$ corresponds to the Rel. 15 $\mathbf{W}_{\text{sub}.1}$ in Type II CSI codebook [REFS], and $\mathbf{B} = \mathbf{W}_{\text{sub}.f}$.

(128) The $\mathbf{C}_{\text{sub}.l} = \{\tilde{\text{over}}(W)\}_{\text{sub}.2}$ matrix consists of all the required linear combination coefficients (e.g., amplitude and phase or real or imaginary). Each reported coefficient $(c_{\text{sub}.l,i,f} = p_{\text{sub}.l,i,f} \phi_{\text{sub}.l,i,f})$ in $\{\tilde{\text{over}}(W)\}_{\text{sub}.2}$ is quantized as amplitude coefficient $(p_{\text{sub}.l,i,f})$ and phase coefficient $(\phi_{\text{sub}.l,i,f})$. In one example, the amplitude coefficient $(p_{\text{sub}.l,i,f})$ is reported using a A-bit amplitude codebook where A belongs to $\{2, 3, 4\}$. If multiple values for A are supported, then one value is configured via higher layer signaling. In another example, the amplitude coefficient $(p_{\text{sub}.l,i,f})$ is reported as $p_{\text{sub}.l,i,f} = p_{\text{sub}.l,i,f,\text{sup}.(1)} p_{\text{sub}.l,i,f,\text{sup}.(2)}$ where $p_{\text{sub}.l,i,f,\text{sup}.(1)}$ is a reference or first amplitude which is reported using a A1-bit amplitude codebook where A1 belongs to $\{2, 3, 4\}$, and $p_{\text{sub}.l,i,f,\text{sup}.(2)}$ is a differential or second amplitude which is reported using a A2-bit amplitude codebook where $A_2 \leq A_1$ belongs to $\{2, 3, 4\}$.

(129) For layer l, let us denote the linear combination (LC) coefficient associated with spatial domain (SD) basis vector (or beam) $i \in \{0, 1, \dots, 2L-1\}$ and frequency domain (FD) basis vector (or beam) $f \in \{0, 1, \dots, M-1\}$ as $c_{\text{sub}.l,i,f}$, and the strongest coefficient as $c_{\text{sub}.l,i^*,f^*}$. The strongest coefficient is reported out of the $K_{\text{sub}.NZ}$ non-zero (NZ) coefficients that is reported using a bitmap, where $K_{\text{sub}.NZ} \leq K_{\text{sub}.0} = \beta \times 2^{LM-1}$ and β is higher layer configured. The remaining $2LM - K_{\text{sub}.NZ}$ coefficients that are not reported by the UE are assumed to be zero. The following quantization scheme is used to quantize/report the $K_{\text{sub}.NZ}$ NZ coefficients.

(130) The UE reports the following for the quantization of the NZ coefficients in $\{\tilde{\text{over}}(W)\}_{\text{sub}.2}$ A X-bit indicator for the strongest coefficient index (i^*, f^*) , where $X = \lceil \log_{\text{sub}.2} K_{\text{sub}.NZ} \rceil$ or $\lceil \log_{\text{sub}.2} 2L \rceil$. Strongest coefficient $c_{\text{sub}.l,i^*,f^*} = 1$ (hence its amplitude/phase are not reported). Two antenna polarization-specific reference amplitudes is used. For the polarization associated with the strongest coefficient $c_{\text{sub}.l,i^*,f^*} = 1$, since the reference amplitude $p_{\text{sub}.l,i^*,f^*,\text{sup}.(1)} = 1$, it is not reported. For the other polarization, reference amplitude $p_{\text{sub}.l,i^*,f^*,\text{sup}.(1)}$ is quantized to 4 bits. The 4-bit amplitude alphabet is

(131) $\{1, (\frac{1}{2})^{\frac{1}{2}}, (\frac{1}{4})^{\frac{1}{2}}, (\frac{1}{8})^{\frac{1}{2}}, \dots, (\frac{1}{2^L})^{\frac{1}{2}}\}$. For $\{c_{\text{sub}.l,i,f}, (i, f) \neq (i^*, f^*)\}$: For each polarization, differential amplitudes $p_{\text{sub}.l,i,f,\text{sup}.(2)}$ of the coefficients calculated relative to the associated polarization-specific reference amplitude and quantized to 3 bits. The 3-bit amplitude alphabet is

(132) $\{1, \frac{1}{\sqrt{2}}, \frac{1}{2}, \frac{1}{2\sqrt{2}}, \frac{1}{4}, \frac{1}{4\sqrt{2}}, \frac{1}{8}, \frac{1}{8\sqrt{2}}\}$. Note: The final quantized amplitude $p_{\text{sub}.l,i,f}$ is given by $p_{\text{sub}.l,i,f,\text{sup}.(1)} \times p_{\text{sub}.l,i,f,\text{sup}.(2)}$. Each phase is quantized to either 8 PSK ($N_{\text{sub}.ph}=8$) or 16 PSK ($N_{\text{sub}.ph}=16$) (which is configurable).

(133) For the polarization $r \in \{0, 1\}$ associated with the strongest coefficient $c_{\text{sub}.l,i^*,f^*}$, we have

$$(134) r^* = \text{Math. } \frac{i^*}{L} \cdot \text{Math.}$$

and the reference amplitude $p_{\text{sub}.l,i^*,f^*,\text{sup}.(1)} = p_{\text{sub}.l,r^*,\text{sup}.(1)} = 1$. For the other polarization $r \in \{0, 1\}$ and $r \neq r^*$, we have

$$(135) r = (\text{Math. } \frac{i^*}{L} \cdot \text{Math.} + 1) \bmod 2$$

and the reference amplitude $p_{\text{sub}.l,i^*,f^*,\text{sup}.(1)} = p_{\text{sub}.l,r,\text{sup}.(1)}$ is quantized (reported) using the 4-bit amplitude codebook mentioned above.

(136) A UE can be configured to report M FD basis vectors. In one example,

$$(137) M = \text{Math. } p \times \frac{N_3}{R} \cdot \text{Math.},$$

where R is higher-layer configured from $\{1, 2\}$ and p is higher-layer configured from

$$(138) \{\frac{1}{4}, \frac{1}{2}\}.$$

In one example, the p value is higher-layer configured for rank 1-2 CSI reporting. For rank >2 (e.g., rank 3-4), the p value (denoted by $v_{\text{sub}.0}$) can be different. In one example, for rank 1-4, $(p, v_{\text{sub}.0})$ is jointly configured from

$$(139) \{(\frac{1}{2}, \frac{1}{4}), (\frac{1}{4}, \frac{1}{4}), (\frac{1}{4}, \frac{1}{8})\},$$

i.e.,

$$(140) M = \text{Math. } p \times \frac{N_3}{R} \cdot \text{Math.}$$

for rank 1-2 and

$$(141) M = \text{Math. } v_0 \times \frac{N_3}{R} \cdot \text{Math.}$$

for rank 3-4. In one example, $N_{\text{sub}.3} = N_{\text{sub}.SB} \times R$ where $N_{\text{sub}.SB}$ is the number of SBs for CQI reporting. In the rest of the disclosure, M is replaced with $M_{\text{sub}.v}$ to show its dependence on the rank value v, hence p is replaced with $p_{\text{sub}.v}$, $v \in \{1, 2\}$ and $v_{\text{sub}.0}$ is replaced with $p_{\text{sub}.v}$, $v \in \{3, 4\}$.

(142) A UE can be configured to report $M_{\text{sub}.v}$ FD basis vectors in one-step from $N_{\text{sub}.3}$ basis vectors freely (independently) for each layer $l \in \{0, 1, \dots, v-1\}$ of a rank v CSI reporting. Alternatively, a UE can be configured to report $M_{\text{sub}.v}$ FD basis vectors in two-step as follows. In step 1, an intermediate set (InS) comprising $N_{\text{sub}.3}' < N_{\text{sub}.3}$ basis vectors is selected/reported, wherein the InS is common for all layers. In step 2, for each layer $l \in \{0, 1, \dots, v-1\}$ of a rank v CSI reporting, $M_{\text{sub}.v}$ FD basis vectors are selected/reported freely (independently) from $N_{\text{sub}.3}'$ basis vectors in the InS.

(143) In one example, one-step method is used when $N_{\text{sub}.3} \leq 19$ and two-step method is used when $N_{\text{sub}.3} > 19$. In one example, $N_{\text{sub}.3}' = \lfloor \alpha M \rfloor$ where $\alpha > 1$ is either fixed (to 2 for example) or configurable.

(144) The codebook parameters used in the DFT based frequency domain compression (Eq. 5) are $(L, p_{\text{sub}.v}$ for $v \in \{1, 2\}$, $p_{\text{sub}.v}$ for $v \in \{3, 4\}$, β , α , $N_{\text{sub}.ph}$). In one example, the set of values for these codebook parameters are as follows. L: the set of values is $\{2, 4\}$ in general, except $L \in \{2, 4, 6\}$ for rank 1-2, 32 CSI-RS antenna ports, and $R=1$.

$$(145) 0(p_v \text{ for } v \in \{1, 2\}, p_v \text{ for } v \in \{3, 4\}) \in \{(\frac{1}{2}, \frac{1}{4}), (\frac{1}{4}, \frac{1}{4}), (\frac{1}{4}, \frac{1}{8})\} \cup \{(\frac{1}{2}, \frac{1}{2}), \frac{3}{4}\} \in \{1.5, 2, 2.5, 3\} N_{ph} \in \{8, 16\}.$$

(146) In another example, the set of values for these codebook parameters are as follows: $\alpha=2$, $N_{\text{sub}.ph}=16$, and as in Table 1, where the values of L, β and $p_{\text{sub}.v}$ are determined by the higher layer parameter paramCombination-r17. In one example, the UE is not expected to be configured with paramCombination-r17 equal to 3, 4, 5, 6, 7, or 8 when $P_{\text{sub}.CSI-RS}=4, 7$ or 8 when number of CSI-RS ports $P_{\text{sub}.CSI-RS} < 32, 7$ or 8 when higher

layer parameter type II-RI-Restriction-r17 is configured with r.sub.i=1 for any i>1, 7 or 8 when R=2.

(147) The bitmap parameter typeII-RI-Restriction-r17 forms the bit sequence r.sub.3, r.sub.2, r.sub.1, r.sub.0 where r.sub.0 is the LSB and r.sub.3 is the MSB. When r.sub.1 is zero, $i \in \{0, 1, \dots, 3\}$, PMI and RI reporting are not allowed to correspond to any precoder associated with v=i+1 layers. The parameter R is configured with the higher-layer parameter numberOfPMISubbandsPerCQISubband-r17. This parameter controls the total number of precoding matrices N.sub.3 indicated by the PMI as a function of the number of subbands in csi-ReportingBand, the subband size configured by the higher-level parameter subbandSize and of the total number of PRBs in the bandwidth part.

(148) TABLE-US-00001 TABLE 1 p.sub.v v paramCombination-r17 $L \in \{1, 2\}$ $\beta \in \{3, 4\}$ $\beta \in \{1, 2, 1/8, 1/4, 1/2, 3/4, 1, 1/8, 1/4, 1/2, 3/4, 1, 1/8, 1/4, 1/2, 3/4\}$

(149) The above-mentioned framework (equation 5) represents the precoding-matrices for multiple (N.sub.3) FD units using a linear combination (double sum) over 2 L SD beams and M.sub.v FD beams. This framework can also be used to represent the precoding-matrices in time domain (TD) by replacing the FD basis matrix W.sub.f with a TD basis matrix W.sub.t, wherein the columns of W.sub.t comprises M.sub.∥ TD beams that represent some form of delays or channel tap locations. Hence, a precoder W.sub.l can be described as follows.

$W = A_{sub.lC.sub.lB.sub.lsup.H} = W_{sub.1} \{ \text{tilde over } (W) \}_{sub.2} W_{sub.tsup.H}$, (Equation 5A)

(150) In one example, the M.sub.v TD beams (representing delays or channel tap locations) are selected from a set of N.sub.3 TD beams, i.e., N.sub.3 corresponds to the maximum number of TD units, where each TD unit corresponds to a delay or channel tap location. In one example, a TD beam corresponds to a single delay or channel tap location. In another example, a TD beam corresponds to multiple delays or channel tap locations. In another example, a TD beam corresponds to a combination of multiple delays or channel tap locations.

(151) The abovementioned framework for CSI reporting based on space-frequency compression (equation 5) or space-time compression (equation 5A) frameworks can be extended to Doppler domain (e.g., for moderate to high mobility UEs). This disclosure focuses on a reference signal burst that can be used to obtain Doppler component(s) of the channel, which can be used to perform Doppler domain compression.

(152) FIG. 13 illustrates an example of a UE configured to receive a burst of non-zero power (NZP) CSI-RS resource(s) 1300 according to embodiments of the present disclosure. The embodiment of the UE configured to receive the burst of NZP CSI-RS resource(s) 1300 illustrated in FIG. 13 is for illustration only. FIG. 13 does not limit the scope of this disclosure to any particular implementation of the UE configured to receive a burst of NZP CSI-RS resource(s) 1300.

(153) In one embodiment A, as shown in FIG. 13, a UE is configured to receive a burst (or occasions) of non-zero power (NZP) CSI-RS resource(s), referred to as CSI-RS burst (or occasions) for brevity, in B time slots, where B≥1. The B time slots can be accordingly to at least one of the following examples. In one example, the B time slots are evenly/uniformly spaced with an inter-slot spacing d. In one example, the B time slots can be non-uniformly spaced with inter-slot spacing e.sub.1=d.sub.1, e.sub.2=d.sub.2−d.sub.1, e.sub.3=d.sub.3−d.sub.2, . . . , so on, where e.sub.i≠e.sub.j for at least one pair (i, j) with i≠j.

(154) The UE receives the CSI-RS burst, estimates the B instances of the DL channel measurements, and uses the channel estimates to obtain the Doppler component(s) of the DL channel. The CSI-RS burst can be linked to (or associated with) a single CSI reporting setting (e.g., via higher layer parameter CSI-ReportConfig), wherein the corresponding CSI report includes an information about the Doppler component(s) of the DL channel.

(155) Let h.sub.t be the DL channel estimate based on the CSI-RS resource(s) received in time slot t∈{0, 1, . . . , B−1}. When the DL channel estimate in slot t is a matrix G.sub.t of size N.sub.Rx×N.sub.Tx×N.sub.Sc, then h.sub.t=vec(G.sub.t), where N.sub.Rx, N.sub.Tx, and N.sub.Sc are number of receive (Rx) antennae at the UE, number of CSI-RS ports measured by the UE, and number of subcarriers in frequency band of the CSI-RS burst, respectively. The notation vec(X) is used to denote the vectorization operation wherein the matrix X is transformed into a vector by concatenating the elements of the matrix in an order, for example, 1.fwdarw.2.fwdarw.3.fwdarw. and so on, implying that the concatenation starts from the first dimension, then moves second dimension, and continues until the last dimension. Let H.sub.B=[h.sub.0 h.sub.1 . . . h.sub.B−1] be a concatenated DL channel. The Doppler component(s) of the DL channel can be obtained based on H.sub.B. For example, H.sub.B can be represented as

$$(156) C^H = \text{Math}_{s=0}^{N-1} C_s^H$$

where $\Phi = [\phi_{sub.0} \phi_{sub.1} \dots \phi_{sub.N-1}]$ is a Doppler domain (DD) basis matrix whose columns comprise basis vectors, $C = [c_{sub.0} c_{sub.1} \dots c_{sub.N-1}]$ is a coefficient matrix whose columns comprise coefficient vectors, and N<B is the number of DD basis vectors. Since the columns of H.sub.B are likely to be correlated, a DD compression can be achieved when the value of N is small (compared to the value of B). In this example, the Doppler component(s) of the channel is represented by the DD basis matrix Φ and the coefficient matrix C.

(157) FIG. 14 illustrates an example of a UE configured to determine a value of N.sub.4 based on the value B in a CSI-RS burst 1400 according to embodiments of the present disclosure. The embodiment of the UE configured to determine a value of N.sub.4 based on the value B in a CSI-RS burst 1400 illustrated in FIG. 14 is for illustration only. FIG. 14 does not limit the scope of this disclosure to any particular implementation of the UE configured to determine a value of N.sub.4 based on the value B in a CSI-RS burst 1400.

(158) Let N.sub.4 be the length of the basis vectors { $\phi_{sub.s}$ }, e.g., each basis vector is a length N.sub.4×1 column vector.

(159) In one example, the UE can be configured to determine/report the CSI report that is valid for C additional time instances or future time instances (as described in embodiment E and F), and hence the UE can perform prediction in C time instances (for example, based on the channel measurements in B time instances), and then determine the DD/TD components of the predicted or/and measured DL channel. In this case, the UE can use the concatenated channel H.sub.B+C=[h.sub.0 h.sub.1 . . . h.sub.B+C−1] or H.sub.C=[h.sub.B h.sub.B+1 . . . h.sub.B+C−1] to determine the DD/TD components of the DL channel. The length of DD/TD basis vectors (N.sub.4) can depend on B (embodiment D) or C (embodiment F) or both B and C (embodiment E).

(160) In one example, the C additional or future time instances belong to a time window that does not include any (CSI-RS) measurement or/and is after the time slot of the CSI reference resource (as described in 5.2.2.5, TS 38.214).

(161) In one embodiment B, a UE is configured to determine a value of N.sub.4 based on the value B (number of CSI-RS instances in a CSI-RS burst) or/and C (number of future/additional time instances), as described above, and determine components across which the DD compression is performed, where each component corresponds to one or multiple time instances within the CSI-RS burst or/and addition/future time instances. In one example, N.sub.4 is fixed (e.g., N.sub.4=B or C or B+C) or configured (e.g., via RRC or MAC CE or DCI) or reported by the UE (as part of the CSI report). In one example, the B or C or B+C time instances can be partitioned into sub-time (ST) units (instances), where each ST unit is defined as (up to) N.sub.ST contiguous time instances in the B or C or B+C time instances. In this example, a component for the DD compression corresponds to a ST unit.

(162) Three examples of the ST units are shown in FIG. 14 for the case of B time instances. In the first example, each ST unit comprises N.sub.ST=1 time instance in the CSI-RS burst. In the second example, each ST unit comprises N.sub.ST=2 contiguous time instances in the CSI-RS burst. In the third example, each ST unit comprises N.sub.ST=4 contiguous time instances in the CSI-RS burst. These examples can be extended to C or B+C time instances in a straight-forward manner.

(163) The value of N.sub.ST can be fixed (e.g., N.sub.ST=1 or 2 or 4) or indicated to the UE (e.g., via higher layer RRC or MAC CE or DCI based signaling) or reported by the UE (e.g., as part of the CSI report). The value of N.sub.ST (fixed or indicated or reported) can be subject to a UE capability reporting. The value of N.sub.ST can also be dependent on the value of B and/or C (e.g., one value for a range of values for B and/or C, and another value for another range of values for B and/or C).

(164) FIG. 15 illustrates an example of a UE configured to determine a value of N.sub.4 with J≥1 CSI-RS bursts that occupy a frequency band and a time span 1500 according to embodiments of the present disclosure. The embodiment of the UE configured to determine a value of N.sub.4 with J≥1

CSI-RS bursts that occupy a frequency band and a time span **1500** illustrated in FIG. 15 is for illustration purposes only. FIG. 15 does not limit the scope of this disclosure to any particular implementation of the UE configured to determine a value of $N_{\text{sub},4}$ with $J \geq 1$ CSI-RS bursts that occupy a frequency band and a time span **1500**.

(165) In one embodiment C, a UE is configured with $J \geq 1$ CSI-RS bursts (as illustrated earlier in the disclosure) that occupy a frequency band and a time span (duration), wherein the frequency band comprises A RBs, and the time span comprises B time instances (of CSI-RS resource(s)) or C or $B+C$ time instances, as described above. When $J > 1$, the A RBs or/and Y time instances (where $Y=B$ or C or $B+C$) can be aggregated across J CSI-RS bursts. In one example, the frequency band equals the CSI reporting band, and the time span equals the number of CSI-RS resource instances (across J CSI-RS bursts) or the time span/window during which the CSI report is expected to be valid, both can be configured to the UE for a CSI reporting, which can be based on the DD compression.

(166) The UE is further configured to partition (divide) the A RBs into subbands (SBs) and/or the Y time instances into sub-times (STs). The partition of A RBs can be based on a SB size value $N_{\text{sub},\text{SB}}$, which can be configured to the UE (cf. Table 5.2.1.4-2 of REFS). The partition of Y time instances can be based either a ST size value $N_{\text{sub},\text{ST}}$ or an r value, as described in this disclosure (cf. embodiment B). An example is illustrated in FIG. 15 for $Y=B$, where $\text{RB}_0, \text{RB}_1, \dots, \text{RB}_{\text{sub},A-1}$ comprise A RBs, $\text{T}_{\text{sub},0}, \text{T}_{\text{sub},1}, \dots, \text{T}_{\text{sub},B-1}$ comprise B time instances, the SB size $N_{\text{sub},\text{SB}}=4$, and the ST size $N_{\text{sub},\text{ST}}=2$.

(167) In one embodiment D, the value of $N_{\text{sub},4}$ depends on B. For example, $N_{\text{sub},4}=B$, or $N_{\text{sub},4}=B \times R_{\text{sub},\text{ST}}$ where $R_{\text{sub},\text{ST}} \geq 1$ (or $R_{\text{sub},\text{ST}} < 1$). So, the DD basis matrix $W_{\text{sub},d}$ comprises N DD basis vectors $\{\phi_{\text{sub},s}\}$, each with length $N_{\text{sub},4} \times 1$. Here, N can be fixed or configured (e.g., via RRC, or MAC CE or DCI). Likewise, B can be fixed or configured (e.g., via RRC, or MAC CE or DCI). When configured, B can be based on the timing (slot indices) of deactivation and activation commands of the SP CSI-RS resource that is configured as the CSI-RS burst, as described in this disclosure. When configured, B can be based on the measurement window of a periodic (P) CSI-RS resource that is configured as the CSI-RS burst, as described in this disclosure.

(168) In one example, the gNB/NW receives the CSI feedback including DD components, where the DD components include the DD basis vectors $\{\phi_{\text{sub},s}\}$. The gNB/NW can use the CSI feedback and predict the precoding matrices for $(C=Q-N_{\text{sub},4})$ future time slots or DD units.

(169) In one example, the gNB/NW determines/constructs basis vectors $\{b_{\text{sub},s}\}$, each with length $Q \times 1$ where $Q > N_{\text{sub},4}$. In one example, $Q=N_{\text{sub},4} \times g$ where g is scaling, e.g., $g=2$.

(170) In one example, when $\phi_{\text{sub},s}$ is a DFT vector with index s, then $b_{\text{sub},s}$ can also be a DFT vector, which can be determined from $\phi_{\text{sub},s}$. In one example, the entries of $\phi_{\text{sub},s}$ can be repeated g times, e.g., $[\phi_{\text{sub},0}, \phi_{\text{sub},0}, \phi_{\text{sub},1}, \phi_{\text{sub},1}, \dots, \phi_{\text{sub},N_{\text{sub},4,\text{sub}}-1}, \phi_{\text{sub},N_{\text{sub},4,\text{sub}}-1}]$ when $g=2$, and the vector $b_{\text{sub},s}$ can be the closest DFT vector of length $Q=gN_{\text{sub},4}$, where the closest DFT vector can be determined based on a metric such as max inner product value, i.e., $\text{abs}(b_{\text{sub},s,\text{sup}} H \phi_{\text{sub},s})$.

(171) In one example, $b_{\text{sub},s}=[\phi_{\text{sub},s}, \dots, \phi_{\text{sub},s}]$ (repeated g times).

(172) In one example, when $g=Q/N_{\text{sub},4}$ is integer, then $b_{\text{sub},s}$ is a DFT vector with index $sg-1$.

(173) In this example, the rest of the components of the codebook are the same as the ones that were reported. For example, when the codebook structure is $W_{\text{sub},1}\{\tilde{\text{over}}(W)\}_{\text{sub},2}(W_{\text{sub},f,\text{Math}}W_{\text{sub},d})_{\text{sup},H}$ or $(W_{\text{sub},f,\text{Math}}W_{\text{sub},1})\{\tilde{\text{over}}(W)\}_{\text{sub},2}W_{\text{sub},d,\text{sup},H}$, and the CSI report includes SD basis $W_{\text{sub},1}$, FD basis $W_{\text{sub},f}$, DD basis $W_{\text{sub},d}$ and coefficient matrix $\{\tilde{\text{over}}(W)\}_{\text{sub},2}$, then the gNB/NW determines/constructs the Q precoders as $W_{\text{sub},1}\{\tilde{\text{over}}(W)\}_{\text{sub},2}(W_{\text{sub},f,\text{Math}}W_{\text{sub},d,\text{pred}})_{\text{sup},H}$ or $(W_{\text{sub},f,\text{Math}}W_{\text{sub},1})W_{\text{sub},2}W_{\text{sub},d,\text{pred},\text{sup},H}$, where $W_{\text{sub},d,\text{pred}}$ is the predicted DD basis matrix comprising N predicted DD basis vectors $\{b_{\text{sub},s}\}$.

(174) In one example, the gNB/NW determines/constructs $W_{\text{sub},2}=\{\tilde{\text{over}}(W)\}_{\text{sub},2}(W_{\text{sub},f,\text{Math}}W_{\text{sub},d})_{\text{sup},H}$ and use it to determine/predict the $W_{\text{sub},2}$ matrices for $(C=Q-N_{\text{sub},4})$ future time slots or DD units. Let $W_{\text{sub},2,\text{pred}}$ is the predicted $W_{\text{sub},2}$ matrix. Then, the gNB/NW determines/constructs the Q precoders as $W_{\text{sub},1}W_{\text{sub},2,\text{pred}}$.

(175) In one example, the gNB/NW determines/constructs precoders $W=W_{\text{sub},1}\{\tilde{\text{over}}(W)\}_{\text{sub},2}(W_{\text{sub},f,\text{Math}}W_{\text{sub},d})_{\text{sup},H}$ or $(W_{\text{sub},f,\text{Math}}W_{\text{sub},1})\{\tilde{\text{over}}(W)\}_{\text{sub},2}W_{\text{sub},d,\text{sup},H}$ and use it to determine/predict the precoding matrices for $(C=Q-N_{\text{sub},4})$ future time slots or DD units.

(176) In one embodiment E, the value of $N_{\text{sub},4}$ depends on B and C (or depends on B, C, and $N_{\text{sub},\text{ST}}$), where B=number of CSI-RS measurement instances (time slots) comprising the CSI-RS burst configured for the CSI reporting including the DD components, C=number of additional time instances or future time slots or prediction window, and $N_{\text{sub},\text{ST}}$ is TD/DD unit size as described above.

(177) In one example, C is length of a CSI reporting/validity window $[l, l+W_{\text{sub},\text{CSI}}-1]$ during which the reported CSI is expected to be valid, i.e., $W_{\text{sub},\text{CSI}}=C$. In one example, $l=n_{\text{sub},\text{ref}}$ and $n_{\text{sub},\text{ref}}$ is the slot index of the CSI reference resource (5.2.2.5, TS 28.214).

(178) In one example, $b \leq B$ and C are included in a CSI reporting/validity window $[l, l+W_{\text{sub},\text{CSI}}-1]$ during which the reported CSI is expected to be valid, i.e., $W_{\text{sub},\text{CSI}}=b+C$. In one example, $l=n_{\text{sub},\text{ref}}-b$ and $n_{\text{sub},\text{ref}}$ is the slot index of the CSI reference resource (5.2.2.5, TS 28.214).

(179) In one example, $N_{\text{sub},4}=T=B+C$. In one example, $N_{\text{sub},4}=T \times R_{\text{sub},\text{ST}}$ where $R_{\text{sub},\text{ST}} \geq 1$ (or $R_{\text{sub},\text{ST}} < 1$).

(180) In one example,

$$(181) N_4 = T_1 = \frac{(B+C)}{N_{\text{ST}}}.$$

In one example, $N_{\text{sub},4}=T_{\text{sub},1} \times R_{\text{sub},\text{ST}}$ where $R_{\text{sub},\text{ST}} \geq 1$ (or $R_{\text{sub},\text{ST}} < 1$).

(182) In one example,

$$(183) N_4 = T_2 = \frac{B}{N_{\text{ST}}} + C.$$

In one example, $N_{\text{sub},4}=T_{\text{sub},2} \times R_{\text{sub},\text{ST}}$ where $R_{\text{sub},\text{ST}} \geq 1$ (or $R_{\text{sub},\text{ST}} < 1$).

(184) In one example,

$$(185) N_4 = T_1 = \text{Math.} \left(\frac{B+C}{N_{\text{ST}}} \right) \text{Math.}.$$

In one example, $N_{\text{sub},4}=T_{\text{sub},1} \times R_{\text{sub},\text{ST}}$ where $R_{\text{sub},\text{ST}} \geq 1$ (or $R_{\text{sub},\text{ST}} < 1$).

(186) In one example,

$$(187) N_4 = T_2 = \text{Math.} \left(\frac{B}{N_{\text{ST}}} + C \right) \text{Math.} \text{ or } \text{Math.} \left(\frac{B}{N_{\text{ST}}} \right) \text{Math.} + C.$$

In one example, $N_{\text{sub},4}=T_{\text{sub},2} \times R_{\text{sub},\text{ST}}$ where $R_{\text{sub},\text{ST}} \geq 1$ (or $R_{\text{sub},\text{ST}} < 1$).

(188) In one example,

$$(189) N_4 = T_1 = \text{Math.} \left(\frac{B+C}{N_{\text{ST}}} \right) \text{Math.}.$$

In one example, $N_{\text{sub},4}=T_{\text{sub},1} \times R_{\text{sub},\text{ST}}$ where $R_{\text{sub},\text{ST}} \geq 1$ (or $R_{\text{sub},\text{ST}} < 1$).

(190) In one example,

$$(191) N_4 = T_2 = \text{Math.} \left(\frac{B}{N_{\text{ST}}} + C \right) \text{Math.} \text{ or } \text{Math.} \left(\frac{B}{N_{\text{ST}}} \right) \text{Math.} + C.$$

In one example, $N_{\text{sub},4}=T_{\text{sub},2} \times R_{\text{sub},\text{ST}}$ where $R_{\text{sub},\text{ST}} \geq 1$ (or $R_{\text{sub},\text{ST}} < 1$).

(192) In one example, there is no measurement RS (e.g., CSI-RS or TRS or SSB) during the C time slots (window). In one example, there can be measurement RS (e.g., CSI-RS or TRS or SSB) during the C time slots (window).

(193) The DD basis matrix $W_{\text{sub},d}$ comprises DD basis vectors $\{\phi_{\text{sub},s}\}$, each with length $N_{\text{sub},4} \times 1$.

(194) In one example, the UE may not or does not need to perform any prediction (for C future slots). The UE can determine the DD basis vectors $\{\phi_{\text{sub},s}\}$ based on the measured DL channel ($H_{\text{sub},\text{meas}}$) during B time slots (measured via the CSI-RS burst configured for CSI reporting), and a portion (or subset of entries) of the DD basis vectors that corresponds to the B CSI-RS measurements. Let $\phi_{\text{sub},s}=[\phi_{\text{sub},s,\text{meas}}, \phi_{\text{sub},s,\text{pred}}]$ be a

DD basis vector with a portion $\phi_{\text{sub.s}}$, means that corresponds to the B CSI-RS measurements and a portion $\phi_{\text{sub.s}}$, pred that corresponds to the C future time slots. Then, $\phi_{\text{sub.s}}$ can be determined based on $\phi_{\text{sub.s}}$, means and H.sub.meas. For instance, the DD basis vectors $\{\phi_{\text{sub.s}}\}$ can be determined based on a metric, e.g., max power or absolute value or max norm value $|\phi_{\text{sub.s}}$, means.sup.HH.sub.meas| or $\text{abs}(\phi_{\text{sub.s}}$, means.sup.HH.sub.meas)

(195) In one example, the UE can use the measured DL channel (H.sub.meas) during B time slots (measured via the CSI-RS burst configured for CSI reporting) to predict/extrapolate the DL channel in C future time slots. The UE can then determine DD basis vectors based on the measured (H.sub.meas) as well as the predicted (H.sub.pred) channels. In this case, the DD basis vectors $\phi_{\text{sub.s}}$ can be determined based on $\text{H.sub.agg} = [\text{H.sub.meas}, \text{H.sub.pred}]$. For instance, the DD basis vectors $\{\phi_{\text{sub.s}}\}$ can be determined based on a metric, e.g., max power or absolute value or max norm value $|\phi_{\text{sub.s}}$, sup.H H.sub.agg| or $\text{abs}(\phi_{\text{sub.s}}$, sup.H H.sub.agg).

(196) In one example, the UE can perform the prediction based on DFT/FFT basis vectors. For example, a DFT/IFFT basis vector $F = [F_{\text{sub.1}}, F_{\text{sub.2}}]$ can be used to determine prediction coefficient vector as $x = F_{\text{sub.1}}$.sup.H $\times$ H.sub.meas, and then use x to predict the channel as $\text{H.sub.pred} = x$.sup.H $\times$ F.sub.2.

(197) The value of B and C can be determined according to at least one of the following examples. In one example, both B and C are fixed. In one example, one of B and C is configured (e.g., via RRC, or MAC CE or DCI), and the other is fixed or determined based on the configured value. For example, B can be configured, and C can be fixed (e.g., $C=10$) or determined based on B (e.g., $C=B$ or $C=t \times B$, and t can be fixed, e.g., 2). In one example, both B and C are configured (e.g., via RRC, or MAC CE or DCI), either via two separate parameters or via a joint parameter. In one example, one of B and C is configured (e.g., via RRC, or MAC CE or DCI), and the other is reported by UE (e.g., via UE capability reporting). For example, B can be configured, and C can be reported.

(198) When both B and C are provided (configured) to UE, then at least one of the following examples is used. In one example, both are configured via the same medium such as RRC, MAC CE or DCI. In one example, one of B and C is configured via RRC, and the other via MAC CE or DCI. In one example, one of B and C is configured via MAC CE, and the other via DCI.

(199) When B and/or C are configured via RRC, then the configuration can be included in or a part of higher layer CSI-ResourceConfig, or higher layer CSI-ReportConfig.

(200) When B and/or C are configured via MAC CE, then the configuration can be included in or a part of CSI trigger state that can be activated/deactivated via MAC CE.

(201) When B and/or C are configured via DCI, then the configuration can be included in or a part of CSI trigger state that can be triggered by DCI.

(202) In one example, one or both of B and N.sub.4 are configured and C is determined based on B and N.sub.4, e.g., $C = N_{\text{sub.4}} - B$. The value of B and N.sub.4 can be determined according to at least one of the following examples. In one example, both B and N.sub.4 are fixed. In one example, one of B and N.sub.4 is configured (e.g., via RRC, or MAC CE or DCI), and the other is fixed or determined based on the configured value. For example, B can be configured, and N.sub.4 can be fixed (e.g., $N_{\text{sub.4}}=10$) or determined based on B (e.g., $N_{\text{sub.4}}=t \times B$, and t can be fixed, e.g., 2). In one example, both B and N.sub.4 are configured (e.g., via RRC, or MAC CE or DCI), either via two separate parameters or via a joint parameter. In one example, one of B and N.sub.4 is configured (e.g., via RRC, or MAC CE or DCI), and the other is reported by UE (e.g., via UE capability reporting). For example, B can be configured, and N.sub.4 can be reported.

(203) When both B and N.sub.4 are provided (configured) to UE, then at least one of the following examples is used. In one example, both are configured via the same medium such as RRC, MAC CE or DCI. In one example, one of B and N.sub.4 is configured via RRC, and the other via MAC CE or DCI. In one example, one of B and N.sub.4 is configured via MAC CE, and the other via DCI.

(204) When B and/or N.sub.4 are configured via RRC, then the configuration can be included in or a part of higher layer CSI-ResourceConfig, or higher layer CSI-ReportConfig.

(205) When B and/or N.sub.4 are configured via MAC CE, then the configuration can be included in or a part of CSI trigger state that can be activated/deactivated via MAC CE.

(206) When B and/or N.sub.4 are configured via DCI, then the configuration can be included in or a part of CSI trigger state that can be triggered by DCI.

(207) The UE is configured to determine/report the CSI report including a PMI, where the PMI indicates SD basis vectors, FD basis vectors, DD basis vectors, and coefficients associated with triples of (SD, FD, DD) basis vectors. The coefficient reporting can be similar to Rel. 16 Type II codebook i.e., an indication of non-zero coefficients (e.g., via bitmap), SCI, reference amplitude, amplitude and phase of NZ coefficients.

(208) The CSI report can also include CQI, where the CQI reporting can be configured to be according to at least one of the following examples. In one example, CQI is reported using both the measurement window or time instances (B), and the prediction or future time instances (C), i.e., $T_{\text{sub.B+C}}$. In one example, CQI is reported using only the measurement window or time instances (B), i.e., $T_{\text{sub.B}}$. In one example, CQI is reported using only the prediction or future time instances (C), i.e., $T_{\text{sub.C}}$.

(209) The number of CQIs reported in TD/DD can be configured to be according to at least one of the following examples. In one example, only one CQI is reported for the entire $T_{\text{sub.x}}$ TD/DD units, where x belongs to {B, C, B+C}. That is, the time granularity of CQI reporting is WT (as described later in this disclosure). When CQI reporting across SBs (the frequency granularity of CQI) is WB, then only one CQI is reported for both CSI reporting band (across SBs) and time window $T_{\text{sub.x}}$. When CQI reporting across SBs (the frequency granularity of CQI) is per SB, then one CQI is reported for each SB, and the reported value remains the same for the time window $T_{\text{sub.x}}$. When the multiple CQIs are reported in a differential manner, then a reference CQI value (for differential CQI reporting) is also reported. In one example, multiple CQIs are reported for the entire $T_{\text{sub.x}}$ TD/DD units, where x belongs to {B, C, B+C}. That is, the time granularity of CQI reporting is ST (as described later in this disclosure). In one example, the number for CQI=2. In one example, the number of CQI is N.sub.4. In one example, the number of CQI is $N_{\text{sub.4}}+1$ (where plus 1 is due for the reference CQI in TD). When CQI reporting across SBs (the frequency granularity of CQI) is WB, then multiple CQIs in TD remains the same for all SBs. When CQI reporting across SBs (the frequency granularity of CQI) is per SB, then multiple CQIs in TD are reported for each SB. When the multiple CQIs are reported in a differential manner, then a reference CQI value (for differential CQI reporting) is also reported or two reference CQI values (one for TD, another for FD) are also reported.

(210) In one embodiment F, the value of N.sub.4 depends on C (or depends on C, and N.sub.ST), where $C = \text{number of additional time instances or future time slots or prediction window}$, and N.sub.ST is TD/DD unit size as described above.

(211) In one example, $N_{\text{sub.4}} = T = C$. In one example, $N_{\text{sub.4}} = T \times R_{\text{sub.ST}}$ where $R_{\text{sub.ST}} \geq 1$ (or $R_{\text{sub.ST}} < 1$).

(212) In one example,

(213) $N_4 = T_1 = \frac{C}{N_{ST}}$.

In one example, $N_{\text{sub.4}} = T_{\text{sub.1}} \times R_{\text{sub.ST}}$ where $R_{\text{sub.ST}} \geq 1$ (or $R_{\text{sub.ST}} < 1$).

(214) In one example,

(215) $N_4 = T_1 = \text{.Math.} \frac{C}{N_{ST}} \text{.Math.}$.

In one example, $N_{\text{sub.4}} = T_{\text{sub.1}} \times R_{\text{sub.ST}}$ where $R_{\text{sub.ST}} \geq 1$ (or $R_{\text{sub.ST}} < 1$).

(216) In one example,

(217) $0N_4 = T_1 = \text{.Math.} \frac{C}{N_{ST}} \text{.Math.}$.

In one example, $N_{\text{sub.4}} = T_{\text{sub.1}} \times R_{\text{sub.ST}}$ where $R_{\text{sub.ST}} > 1$ (or $R_{\text{sub.ST}} < 1$).

(218) In one example, there is no measurement RS (e.g., CSI-RS or TRS or SSB) during the C time slots (window). In one example, there can be

measured RS (e.g., CSI-RS or SSB) during the C time slots (window).

(219) The DD basis matrix $W_{sub,d}$ comprises DD basis vectors $\{\phi_{sub,s}\}$, each with length $N_{sub,4} \times 1$.

(220) In one example, the UE may not or does not need to perform any prediction (for C future slots). The UE can determine the DD basis vectors $\{\phi_{sub,s}\}$ based on the measured DL channel ($H_{sub,meas}$) during B time slots (measured via the CSI-RS burst configured for CSI reporting).

(221) In one example, the UE can use the measured DL channel ($H_{sub,meas}$) during B time slots (measured via the CSI-RS burst configured for CSI reporting) to predict/extrapolate the DL channel in C future time slots. The UE can then determine DD basis vectors based on the predicted ($H_{sub,pred}$) channels. For instance, the DD basis vectors $\{\phi_{sub,s}\}$ can be determined based on a metric, e.g., max power or absolute value or max norm value $|\phi_{sub,s}^{sup.H} H_{sub,pred}|$ or $abs(\phi_{sub,s}^{sup.H} H_{sub,pred})$.

(222) In one example, the UE can perform the prediction based on DFT/FFT basis vectors. For example, a DFT/IFFT basis vector $F=[F_{sub,1}, F_{sub,2}]$ can be used to determine prediction coefficient vector as $x=F_{sub,1}^{sup.H} \times H_{sub,meas}$, and then use x to predict the channel as $H_{sub,pred}=x^{sup.H} \times F_{sub,2}$.

(223) The value of C can be determined according to at least one of the following examples. In one example, C is fixed. In one example, C is configured (e.g., via RRC, or MAC CE or DCI). In one example, C is reported by UE (e.g., via UE capability reporting).

(224) When C is configured via RRC, then the configuration can be included in or a part of higher layer CSI-ResourceConfig, or higher layer CSI-ReportConfig.

(225) When C is configured via MAC CE, then the configuration can be included in or a part of CSI trigger state that can be activated/deactivated via MAC CE.

(226) When C is configured via DCI, then the configuration can be included in or a part of CSI trigger state that can be triggered by DCI.

(227) In one example, $N_{sub,4}$ is configured and C is determined based on $N_{sub,4}$. The value of C and $N_{sub,4}$ can be determined according to at least one of the following examples. In one example, both C and $N_{sub,4}$ are fixed. In one example, one of C and $N_{sub,4}$ is configured (e.g., via RRC, or MAC CE or DCI), and the other is fixed or determined based on the configured value. For example, C can be configured, and $N_{sub,4}$ can be fixed (e.g., $N_{sub,4}=10$) or determined based on C (e.g., $N_{sub,4}=t \times C$, and t can be fixed, e.g., 2).

(228) In one example, both C and $N_{sub,4}$ are configured (e.g., via RRC, or MAC CE or DCI), either via two separate parameters or via a joint parameter.

(229) In one example, one of C and $N_{sub,4}$ is configured (e.g., via RRC, or MAC CE or DCI), and the other is reported by UE (e.g., via UE capability reporting). For example, C can be configured, and $N_{sub,4}$ can be reported.

(230) When both C and $N_{sub,4}$ are provided (configured) to UE, then at least one of the following examples is used. In one example, both are configured via the same medium such as RRC, MAC CE or DCI. In one example, one of C and $N_{sub,4}$ is configured via RRC, and the other via MAC CE or DCI. In one example, one of C and $N_{sub,4}$ is configured via MAC CE, and the other via DCI.

(231) When C or/and $N_{sub,4}$ are configured via RRC, then the configuration can be included in or a part of higher layer CSI-ResourceConfig, or higher layer CSI-ReportConfig.

(232) When C or/and $N_{sub,4}$ are configured via MAC CE, then the configuration can be included in or a part of CSI trigger state that can be activated/deactivated via MAC CE.

(233) When C or/and $N_{sub,4}$ are configured via DCI, then the configuration can be included in or a part of CSI trigger state that can be triggered by DCI.

(234) The UE is configured to determine/report the CSI report including a PMI, where the PMI indicates SD basis vectors, FD basis vectors, DD basis vectors, and coefficients associated with triples of (SD, FD, DD) basis vectors. The coefficient reporting can be similar to Rel. 16 Type II codebook i.e., an indication of non-zero coefficients (e.g., via bitmap), SCI, reference amplitude, amplitude and phase of NZ coefficients.

(235) In the rest of the disclosure, the signaling/configuration related to the (time or/and frequency) granularity of (components of) the CSI reporting (based on the DD compression) is discussed, where the 'granularity' of a quantity refers to the number of values of the quantity is reported in the CSI reported. For example, when the quantity is CQI, then the granularity of the CQI can correspond to one (i.e., only one CQI value is reported) or multiple (i.e., multiple CQI values are reported).

(236) In one embodiment I.1, the frequency granularity (i.e., the granularity in FD) of a component, say X, of the CSI report is determined/configured according to at least one of the following examples.

(237) In one example I.1.1, the frequency granularity is wideband (WB), i.e., one (single) wideband X is reported for each codeword for the entire CSI reporting band (comprising A RBs).

(238) In one example I.1.2, the frequency granularity is subband (SB) i.e., multiple X values are reported, one for each codeword is reported for each SB in the CSI reporting band.

(239) In one example I.1.3, the frequency granularity is subband (SB) w.r.t. to a WB reference, i.e., it is a combination of WB and SB (or differential SB w.r.t. WB). As reference, one (single) wideband X reported for each codeword for the entire CSI reporting band. As differential, one X for each codeword is reported for each subband in the CSI reporting band. For each SB index s, a SB differential X is defined as: SB offset level (s)=SB X index (s)-WB X index. An example of the mapping from the 2-bit SB differential X values to the offset level is shown in Table 2.

(240) TABLE-US-00002 TABLE 2 Mapping and differential X value to offset level differential X value Offset level 0 0 1 1 2 ≥ 3 -1

(241) In one example, the component X in this embodiment can only be CQI. In one example, the component X in this embodiment can be CQI or PMI. In one example, the component X in this embodiment can be CQI or RI. In one example, the component X in this embodiment can be CQI or PMI or RI.

(242) In one example, the frequency granularity of the component X is determined by the UE and this information is reported by the UE, e.g., as part of the CSI report or UE capability information. In one example, the frequency granularity of the component X is configured to the UE, e.g., via RRC, MAC-CE, or DCI, and the configuration can be subject to the UE capability reported by the UE.

(243) The partitioning of A RBs into SBs is performed when the frequency granularity is according to Example I.1.2 or I.1.3. For some components X (e.g., PMI), each SB can be partitioned further into (up to) R parts, similar to the PMI reporting based on the Rel. 16 enhanced Type II codebook. The value of R can belong to $\{1, 2, N_{sub,SB}\}$. The value of R can be configured (e.g., via higher layer), which can be subject to the UE capability reporting. In one example, the support for R=1 is mandatory (hence, doesn't require additional signaling from the UE), and the support of R>1 is optional (hence, requires additional signaling if the UE supports it). Examples of two R values, R=1, 2 are illustrated in FIG. 15, where when R=1, SBs are not partitioned (i.e., 4 RBs form a SB), and when R=2, each SB is partitioned into two parts (i.e., 4 RBs form a SB, which is partitioned into 2 parts, each with 2 RBs). When R>1, a total of $N_{sub,3}$ quantities of X are reported (e.g., via the PMI when X=PMI), one for each FD unit/component, where $N_{sub,3}=R \times N_{sub,SB}^{sup.tot}$ when each SB is partitioned into R parts, and $N_{sub,3}=R \times (N_{sub,SB}^{sup.tot}-2)+P+Q$ in general, where $N_{sub,SB}^{sup.tot}$ is the total number of SBs, P is number of parts of the first SB, Q is number of parts of the last SB, and $P, Q \in \{1, \dots, R\}$.

(244) In one embodiment I.2, the time granularity (i.e., the granularity in TD) of a component, say X, of the CSI report is determined/configured according to at least one of the following examples.

(245) In one example I.2.1, the time granularity is wide-time (WT), i.e., one (single) wide-time X is reported for each codeword for the entire time span (duration) comprising B or C or B+C time instances.

(246) In one example I.2.2, the time granularity is subtime (ST) i.e., multiple X values are reported, one for each codeword is reported for each ST in the entire time span.

(247) In one example I.2.3, the time granularity of subtime (ST) w.r.t. to a WT reference, i.e., it is a combination of WT and ST (or differential ST w.r.t. WT). As reference, one (single) wide time X reported for each codeword for the entire time span. As differential, one X for each codeword is reported for each subtime in the entire time span. For each ST index t , a ST differential X is defined as: ST offset level (t)=ST X index (t)-WT X index. An example of the mapping from the 2-bit ST differential X values to the offset level is shown in Table 2.

(248) In one example, the component X in this embodiment can only be CQI. In one example, the component X in this embodiment can be CQI or PMI. In one example, the component X in this embodiment can be CQI or RI. In one example, the component X in this embodiment can be CQI or PMI or RI.

(249) In one example, the time granularity of the component X is determined by the UE and this information is reported by the UE, e.g., as part of the CSI report or UE capability information. In one example, the time granularity of the component X is configured to the UE, e.g., via RRC, MAC-CE, or DCI, and the configuration can be subject to the UE capability reported by the UE.

(250) The partitioning of B or C or B+C time instances into STs is performed when the time granularity is according to Example I.2.2 or I.2.3. For some components X (e.g., PMI), each ST can be partitioned further into (up to) R.sub.ST parts, similar to the PMI reporting based on the Rel. 16 enhanced Type II codebook. The value of R.sub.ST can belong to {1, 2, N.sub.ST}. The value of R.sub.ST can be configured (e.g., via higher layer), which can be subject to the UE capability reporting. In one example, the support for R.sub.ST=1 is mandatory (hence, doesn't require additional signaling from the UE), and the support of R.sub.ST>1 is optional (hence, requires additional signaling if the UE supports it). Examples of two R.sub.ST values, R.sub.ST=1, 2 are illustrated in FIG. 15 for B time instances, where when R.sub.ST=1, STs are not partitioned (i.e., 2 time instances form a ST), and when R.sub.ST=2, each ST is partitioned into two parts (2 time instances form a ST, which is partitioned into 2 parts, each with 1 time instance). When R.sub.ST>1, a total of N.sub.4 quantities of X are reported (via the PMI when X=PMI), one for each ST unit/component, where $N_{sub.4}=R_{sub.ST} \times N_{sub.ST.sup.tot}$ when each ST is partitioned into R.sub.ST parts, and $N_{sub.4}=R_{sub.ST} \times (N_{sub.ST.sup.tot}-2) + P_{sub.ST} + Q_{sub.ST}$ in general, where N.sub.ST.sup.tot is the total number of STs, P.sub.ST is number of parts of the first ST, Q.sub.ST is number of parts of the last ST, and $P_{sub.ST}, Q_{sub.ST} \in \{1, \dots, R_{sub.ST}\}$.

(251) In one embodiment I.3, the frequency and time granularities (i.e., granularity in FD and TD) of a component, say X, of the CSI report is according to at least one of the following examples.

(252) In one example I.3.1, both frequency and time granularities are fixed, hence there is no need for any signaling or configuration. In particular, the frequency and time granularities are fixed to example I.1.a and example I.2.b, respectively, where (a, b) is one of (1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), and (3,3).

(253) In one example I.3.2, the frequency granularity is signaled/configured (e.g., RRC, MAC-CE, or DCI), and the time granularity is fixed. In particular, the frequency granularity is signaled/configured from example I.1.1 through I.1.3, and the time granularity is fixed to one of example I.2.1 through I.2.3.

(254) In one example I.3.3, the time granularity is signaled/configured (e.g., RRC, MAC-CE, or DCI), and the frequency granularity is fixed. In particular, the time granularity is signaled/configured from example I.2.1 through I.2.3, and the frequency granularity is fixed to one of example I.1.1 through I.1.3.

(255) In one example I.3.4, both frequency and time granularities are signaled/configured (e.g., RRC, MAC-CE, or DCI). In particular, the frequency granularity is signaled/configured from example I.1.1 through I.1.3, and the time granularity is signaled/configured from example I.2.1 through I.2.3. The signaling/configuration of the frequency and time granularities can be separate (i.e., via two separate parameters or configurations), or joint (i.e., via one parameter or configuration).

(256) In one example, the component X in this embodiment can only be CQI. In one example, the component X in this embodiment can be CQI or PMI. In one example, the component X in this embodiment can be CQI or RI. In one example, the component X in this embodiment can be CQI or PMI or RI.

(257) In one example, the frequency and time granularities of the component X are determined by the UE and this information is reported by the UE, e.g., as part of the CSI report or UE capability information. In one example, the frequency and time granularities of the component X are configured to the UE, e.g., via RRC, MAC-CE, or DCI, and the configuration can be subject to the UE capability reported by the UE. In one example, one of the frequency and time granularities is determined by the UE and this information is reported by the UE, e.g., as part of the CSI report or UE capability information, and the other of the frequency and time granularities is configured to the UE, e.g., via RRC, MAC-CE, or DCI.

(258) The partitioning of A RB s into SB s is performed when the frequency granularity is according to Example I.1.2 or I.1.3, the details as in embodiment I.1. Likewise, the partitioning of B or C or B+C time instances into STs is performed when the time granularity is according to Example I.2.2 or I.2.3, the details as in embodiment I.2.

(259) FIG. 16 illustrates an example of the time and frequency granularities of reporting X **1600** according to embodiments of the present disclosure. The embodiment of the time and frequency granularities of reporting X **1600** illustrated in FIG. 16 is for illustration only. FIG. 16 does not limit the scope of this disclosure to any particular implementation of the time and frequency granularities of reporting X **1600**.

(260) An illustration of the time and frequency granularities of reporting X is shown in FIG. 16, wherein four examples are shown: when the reporting format is (WB, WT), only one value for X is reported for the whole CSI band and for entire time span; when the reporting format is (WB, ST), four values for X are reported, each value is for the whole CSI band and a specific ST; when the reporting format is (SB, WT), 8 values for X are reported, each value is for a specific SB and the entire time span; when the reporting format is (SB, ST), 32 values for X are reported, each value is for a specific SB and a specific ST.

(261) In one embodiment I.4, the frequency and time granularities of two components, say X.sub.1 and X.sub.2, of the CSI report is according to at least one of the following examples.

(262) In one example I.4.1, the frequency and time granularities for both X.sub.1 and X.sub.2 are fixed, hence there is no need for any signaling or configuration. In particular, the frequency and time granularities for both X.sub.1 and X.sub.2 are fixed according to example I.3.1 with (a, b)=(a.sub.1, b.sub.1) and (a, b)=(a.sub.2, b.sub.2) for X.sub.1 and X.sub.2, respectively. In one example, (a.sub.1, b.sub.1)=(a.sub.2, b.sub.2). In one example, a.sub.1=a.sub.2 but b.sub.1 and b.sub.2 can be the same or different. In one example, b.sub.1=b.sub.2 but a.sub.1 and a.sub.2 can be the same or different. In one example, both a.sub.1 and a.sub.2, and b.sub.1 and b.sub.2 can be the same or different.

(263) In one example I.4.2, the frequency and time granularities for X.sub.1 are signaled/configured (e.g., RRC, MAC-CE, or DCI), and the frequency and time granularities for X.sub.2 are fixed (i.e., signaling/configuration is only for X.sub.1). In particular, the frequency and time granularities for X.sub.1 is signaled/configured from example I.3.2 through I.3.4, and the frequency and time granularities for X.sub.2 are fixed according to example I.3.1.

(264) In one example I.4.3, the frequency and time granularities for X.sub.2 are signaled/configured (e.g., RRC, MAC-CE, or DCI), and the frequency and time granularities for X.sub.1 are fixed (i.e., signaling/configuration is only for X.sub.2). In particular, the frequency and time granularities for X.sub.2 is signaled/configured from example I.3.2 through I.3.4, and the frequency and time granularities for X.sub.1 are fixed according to example I.3.1.

(265) In one example I.4.4, the frequency and time granularities for both X.sub.1 and X.sub.2 are signaled/configured (e.g., RRC, MAC-CE, or DCI). In particular, the frequency and time granularities for both X.sub.1 and X.sub.2 are signaled/configured from example I.3.2 through I.3.4. In one example, the frequency and time granularities for both X.sub.1 and X.sub.2 are the same, hence, the one/same example from example I.3.2 through I.3.4 is used. In one example, the frequency and time granularities for both X.sub.1 and X.sub.2 can be the same or different, hence two examples, each from example I.3.2 through I.3.4 is used, one for X.sub.1 and another for X.sub.2. Also, the signaling/configuration of the frequency

and time granularities for X.sub.1 and X.sub.2 can be separate (i.e., via two separate parameters or configurations), or joint (i.e., via one parameter or configuration).

(266) In one example, the two components (X.sub.1 and X.sub.2) in this embodiment can only be X.sub.1=PMI and X.sub.2=CQI. In one example, two components (X.sub.1 and X.sub.2) in this embodiment can be (X.sub.1, X.sub.2)=(PMI, CQI), or (PMI, RI), or (RI, CQI).

(267) In one example, the frequency and time granularities of the components X.sub.1 and X.sub.2 are determined by the UE and this information is reported by the UE, e.g., as part of the CSI report or UE capability information. In one example, the frequency and time granularities of the components X.sub.1 and X.sub.2 are configured to the UE, e.g., via RRC, MAC-CE, or DCI, and the configuration can be subject to the UE capability reported by the UE. In one example, one of the frequency and time granularities of component X.sub.i (i=1, 2) is determined by the UE and this information is reported by the UE, e.g., as part of the CSI report or UE capability information, and the other of the frequency and time granularities of component X.sub.i (i=1, 2) is configured to the UE, e.g., via RRC, MAC-CE, or DCI.

(268) In one embodiment 1.5, when the frequency and time granularities of CQI reporting is signaled/configured (e.g., RRC, MAC-CE, or DCI) as described in example 1.3.4, the signaling/configuration can be according to at least one of the following examples.

(269) In one example 1.5.1, the signaling is separate via two separate parameters, for example, using a first parameter `cqiFormatIndicator` or `cqiFrequencyFormatIndicator` indicating a value from {wideband, subband} for frequency granularity, and using a second parameter `cqiFormatIndicator2` or `cqiTimeFormatIndicator` indicating a value from {wide-time, sub-time} for time granularity.

(270) In one example 1.5.2, the signaling is joint via one parameter, for example, using a joint parameter `cqiFormatIndicator` indicating a pair of values from {(wideband, wide-time), (wideband, sub-time), (subband, wide-time), (subband, sub-time)} for (frequency, time) granularity pair.

(271) In one embodiment 1.6, when the frequency and time granularities of PMI reporting is signaled/configured (e.g., RRC, MAC-CE, or DCI) as described in example 1.3.4, the signaling/configuration can be according to at least one of the following examples.

(272) In one example 1.6.1, the signaling is separate via two separate parameters, for example, using a first parameter `pmiFormatIndicator` or `pmiFrequencyFormatIndicator` indicating a value from {wideband, subband} for frequency granularity, and using a second parameter `pmiFormatIndicator2` or `pmiTimeFormatIndicator` indicating a value from {wide-time, sub-time} for time granularity.

(273) In one example 1.6.2, the signaling is joint via one parameter, for example, using a joint parameter `pmiFormatIndicator` indicating a pair of values from {(wideband, wide-time), (wideband, sub-time), (subband, wide-time), (subband, sub-time)} for (frequency, time) granularity pair.

(274) In one embodiment 1.7, when the frequency and time granularities of both CQI and PMI reporting are signaled/configured (e.g., RRC, MAC-CE, or DCI) as described in example 1.3.4, the signaling/configuration can be according to at least one of the following examples.

(275) In one example 1.7.1, the signaling/configuration is common for both PMI and CQI and indicates the same time and frequency granularities for PMI and CQI. In one example 1.7.1.1, the signaling is separate via two separate parameters, for example, using a first parameter `FormatIndicator` or `frequencyFormatIndicator` indicating a value from {wideband, subband} for frequency granularity, and using a second parameter `formatIndicator2` or `timeFormatIndicator` indicating a value from {wide-time, sub-time} for time granularity. In one example 1.7.1.2, the signaling is joint via one parameter, for example, using a joint parameter `formatIndicator` indicating a pair of values from {(wideband, wide-time), (wideband, sub-time), (subband, wide-time), (subband, sub-time)} for (frequency, time) granularity pair.

(276) In one example 1.7.2, the signaling/configuration of the time granularity is common for PMI and CQI and indicates the same time granularity for PMI and CQI; and the signaling/configuration of the frequency granularity is separate for PMI and CQI. In one example 1.7.2.1, the signaling is via 3 parameters (p1,p2,p3) p1=`timeFormatIndicator` indicating a value from {wide-time, sub-time} for time granularity, p2=`pmiFrequencyFormatIndicator` indicating a value from {wideband, subband} for frequency granularity of PMI, p3=`cqiFrequencyFormatIndicator` indicating a value from {wideband, subband} for frequency granularity of CQI. In one example 1.7.2.2, the signaling is via 2 parameters (p1,p2) which are according to one of the following examples. In one example, p1 indicates a pair (`timeFormatIndicator`, `cqiFrequencyFormatIndicator`), and p2=`pmiFrequencyFormatIndicator`. In one example, p1 indicates a pair (`timeFormatIndicator`, `pmiFrequencyFormatIndicator`), and p2=`cqiFrequencyFormatIndicator`. In one example, p1 indicates a pair (`pmiFrequencyFormatIndicator`, `cqiFrequencyFormatIndicator`), and p3=`timeFormatIndicator`. In one example 1.7.2.3, the signaling is via 1 parameter p indicating a triple (`timeFormatIndicator`, `pmiFrequencyFormatIndicator`, `cqiFrequencyFormatIndicator`).

(277) In one example 1.7.3, the signaling/configuration of the frequency granularity is common for PMI and CQI and indicates the same frequency granularity for PMI and CQI; and the signaling/configuration of the time granularity is separate for PMI and CQI. In one example 1.7.3.1, the signaling is via 3 parameters (p1,p2,p3) p1=`frequencyFormatIndicator` indicating a value from {wideband, subband} for frequency granularity, p2=`pmiTimeFormatIndicator` indicating a value from {wide-time, sub-time} for time granularity of PMI, p3=`cqiTimeFormatIndicator` indicating a value from {wide-time, sub-time} for time granularity of CQI. In one example 1.7.3.2, the signaling is via 2 parameters (p1,p2) which are according to one of the following examples. In one example, p1 indicates a pair (`frequencyFormatIndicator`, `pmiTimeFormatIndicator`), and p2=`cqiTimeFormatIndicator`. In one example, p1 indicates a pair (`frequencyFormatIndicator`, `cqiTimeFormatIndicator`), and p2=`pmiTimeFormatIndicator`. In one example, p1 indicates a pair (`pmiTimeFormatIndicator`, `cqiTimeFormatIndicator`), and p3=`frequencyFormatIndicator`. In one example 1.7.3.3, the signaling is via 1 parameter p indicating a triple (`frequencyFormatIndicator`, `pmiTimeFormatIndicator`, `cqiTimeFormatIndicator`).

(278) In one example 1.7.4, the signaling/configuration is separate for PMI and CQI. For CQI, it is according to one of example 1.5.1 and 1.5.2, and for PMI, it is according to one of example 1.6.1 and 1.6.2. Let a=`pmiTimeFormatIndicator` indicating a value from {wide-time, sub-time} for time granularity of PMI, b=`cqiTimeFormatIndicator` indicating a value from {wide-time, sub-time} for time granularity of CQI, c=`pmiFrequencyFormatIndicator` indicating a value from {wideband, subband} for frequency granularity of PMI, d=`cqiFrequencyFormatIndicator` indicating a value from {wideband, subband} for frequency granularity of CQI. In one example 1.7.4.1, the signaling is via 4 parameters (p1,p2,p3,p4): p1=a, p2=b, p3=c, p4=d. In one example 1.7.4.2, the signaling is via 3 parameters (p1,p2,p3) parameters, which are according to one of the following examples. In one example, p1 indicates a pair (a,b), p2=c, and p3=d. In one example, p1 indicates a pair (a,c), p2=b, and p3=d. In one example, p1 indicates a pair (a,d), p2=b, and p3=c. In one example, p1 indicates a pair (b,c), p2=a, and p3=d. In one example, p1 indicates a pair (b,d), p2=a, and p3=c. In one example, p1 indicates a pair (c,d), p2=a, and p3=b. In one example 1.7.4.3, the signaling is via 2 parameters (p1,p2) parameters, which are according to one of the following examples. In one example, p1 indicates a triple (a,b,c), and p2=d. In one example, p1 indicates a triple (a,b,d), and p2=c. In one example, p1 indicates a triple (b,c,d), and p2=a. In one example, p1 indicates a triple (a,c,d), and p2=b. In one example 1.7.4.4, the signaling is via p indicates a tuple (a,b,c,d).

(279) The frequency and time granularities are not separate (decoupled), they are rather coupled, e.g., as a two-dimensional (2D) granularity, namely frequency-time (F-T) or time-frequency (T-F) granularity. The CSI reporting band comprising A RBs and the time span (duration) comprising B time instances comprise the (2D) F-T measurement window.

(280) FIG. 17 illustrates an example of the F-T granularity 1700 according to embodiments of the present disclosure. The embodiment of the F-T granularity 1700 illustrated in FIG. 17 is for illustration only. FIG. 17 does not limit the scope of this disclosure to any particular implementation of F-T granularity 1700.

(281) In one embodiment II.1, the F-T granularity of a component, say X, of the CSI report is according to at least one of the following examples.

(282) In one example II.1.1, the F-T granularity is widebandAndTime (WBT), i.e., one (single) X is reported for each codeword for the entire 2D F-T measurement window.

(283) In one example II.1.2, the T-F granularity is subbandAndTime (SBT) i.e., multiple X values are reported, one for each codeword is reported for each SBT in the 2D F-T measurement window.

(284) In one example II.1.3, the F-T granularity is subbandAndTime (SBT) w.r.t. to a WBT reference, i.e., a combination of WBT and SBT (or differential SBT w.r.t. WBT). As reference, one (single) wideband X reported for each codeword for the entire 2D F-T measurement window. As differential, one X for each codeword is reported for each SBT in the 2D F-T measurement window. For each SBT index s, a SBT differential X is defined as: SBT offset level (s)=SBT X index (s)—WBT X index. An example of the mapping from the 2-bit SBT differential X values to the offset level is shown in Table 2.

(285) In examples II.1.2 and II.1.3, the F-T measurement window is partitioned into multiple units, say subbandAndTime (SBT) units. The size of a SBT unit depends on the number of RBs and time instances comprising an SBT. An illustration of the F-T granularity is shown in FIG. 17, wherein six examples are shown: when the reporting format is WBT, only one value for X is reported for the entire 2D F-T measurement window; when the reporting format is SBT, multiple values for X are reported, one for each SBT. Five examples of SBTs with different sizes are shown.

(286) In one example, an SBT unit corresponds to a regular shape (e.g., as shown in FIG. 17). The size of an SBT unit can be fixed, or configured (e.g., via RRC, MAC-CE, or DCI), or reported by the UE (e.g., as part of CSI report or UE capability reporting).

(287) In one example, an SBT unit corresponds to a pattern, which can be irregular in shape. The SBT pattern can be fixed, or configured (e.g., via RRC, MAC-CE, or DCI), or reported by the UE (e.g., as part of CSI report or UE capability reporting).

(288) In one example, an SBT unit corresponds to a regular shape or a pattern (which can be irregular in shape). The size or pattern of an SBT unit can be fixed, or configured (e.g., via RRC, MAC-CE, or DCI), or reported by the UE (e.g., as part of CSI report or UE capability reporting).

(289) In one embodiment II.2, the F-T granularity of a component, say X, of the CSI report is according to at least one of the following examples.

(290) In one example II.2.1, the F-T granularity is fixed, hence there is no need for any signaling or configuration. In particular, the F-T granularity is fixed to example II.1.a, where a is one of 1, 2, or 3.

(291) In one example II.2.2, the F-T granularity is signaled/configured (e.g., RRC, MAC-CE, or DCI). In particular, it is signaled/configured from example II.1.1 through II.1.3.

(292) In one example, the component X in this embodiment can only be CQI. In one example, the component X in this embodiment can be CQI or PMI. In one example, the component X in this embodiment can be CQI or RI. In one example, the component X in this embodiment can be CQI or PMI or RI.

(293) In one example, the F-T granularity of the component X is determined by the UE and this information is reported by the UE, e.g., as part of the CSI report or UE capability information. In one example, the F-T granularity of the component X is configured to the UE, e.g., via RRC, MAC-CE, or DCI, and the configuration can be subject to the UE capability reported by the UE.

(294) In one embodiment II.3, the F-T granularities of two components, say X.sub.1 and X.sub.2, of the CSI report is according to at least one of the following examples.

(295) In one example II.3.1, the F-T granularities for both X.sub.1 and X.sub.2 are fixed, hence there is no need for any signaling or configuration. In particular, the F-T granularities for both X.sub.1 and X.sub.2 are fixed according to example II.2.1 with a=a.sub.1 and a=a.sub.2 for X.sub.1 and X.sub.2, respectively. In one example, a.sub.1=a.sub.2. In one example, a.sub.1 and a.sub.2 can be the same or different.

(296) In one example II.3.2, the F-T granularity for X.sub.1 is signaled/configured (e.g., RRC, MAC-CE, or DCI), and the F-T granularity for X.sub.2 are fixed (i.e., signaling/configuration is only for X.sub.1). In particular, the F-T granularity for X.sub.1 is signaled/configured according to example II.2.2, and the F-T granularity for X.sub.2 is fixed according to example II.2.1.

(297) In one example II.3.3, the F-T granularity for X.sub.2 is signaled/configured (e.g., RRC, MAC-CE, or DCI), and the F-T granularity for X.sub.1 are fixed (i.e., signaling/configuration is only for X.sub.2). In particular, the F-T granularity for X.sub.2 is signaled/configured according to example II.2.2, and the F-T granularity for X.sub.1 is fixed according to example II.2.1.

(298) In one example II.3.4, the F-T granularities for both X.sub.1 and X.sub.2 are signaled/configured (e.g., RRC, MAC-CE, or DCI). In particular, the F-T granularities for both X.sub.1 and X.sub.2 are signaled/configured according to example II.2.2. In one example, the F-T granularities for both X.sub.1 and X.sub.2 are the same, hence, the one/same/single configuration is used/sufficient. In one example, the F-T granularities for both X.sub.1 and X.sub.2 can be the same or different, hence two configurations, one for X.sub.1 and another for X.sub.2, is used/needed. Also, the signaling/configuration of the F-T granularities for X.sub.1 and X.sub.2 can be separate (i.e., via two separate parameters or configurations), or joint (i.e., via one parameter or configuration).

(299) In one example, the two components (X.sub.1 and X.sub.2) in this embodiment can only be X.sub.1=PMI and X.sub.2=CQI. In one example, two components (X.sub.1 and X.sub.2) in this embodiment can be (X.sub.1, X.sub.2)=(PMI, CQI), or (PMI, RI), or (RI, CQI).

(300) In one example, the F-T granularity of the components X.sub.1 and X.sub.2 are determined by the UE and this information is reported by the UE, e.g., as part of the CSI report or UE capability information. In one example, the F-T granularity of the components X.sub.1 and X.sub.2 are configured to the UE, e.g., via RRC, MAC-CE, or DCI, and the configuration can be subject to the UE capability reported by the UE. In one example, the F-T granularity of one of components X.sub.i (i=1, 2) is determined by the UE and this information is reported by the UE, e.g., as part of the CSI report or UE capability information, and the F-T granularity of another of components X.sub.i (i=1, 2) is configured to the UE, e.g., via RRC, MAC-CE, or DCI.

(301) In one embodiment II.4, the F-T granularities of at least one of CQI and PMI is signaled/configured according to at least one of the following examples.

(302) In one example II.4.1, when the F-T granularity of CQI reporting is signaled/configured (e.g., RRC, MAC-CE, or DCI) as described in example II.2.2, the signaling/configuration can be via a parameter, for example, parameter cqiFormatIndicator, indicating a value from {WBT, SBT}.

(303) In one example II.4.2, when the F-T granularity of PMI reporting is signaled/configured (e.g., RRC, MAC-CE, or DCI) as described in example II.2.2, the signaling/configuration can be via a parameter, for example, parameter pmiFormatIndicator, indicating a value from {WBT, SBT}.

(304) In one example II.4.3, when the F-T granularities of both CQI and PMI reporting are signaled/configured (e.g., RRC, MAC-CE, or DCI) as described in example II.2.2, the signaling/configuration can be according to at least one of the following examples. In one example II.4.3.1, the F-T granularities of CQI and PMI are the same, hence the signaling/configuration is common for both PMI and CQI and indicates the same/one/single F-T granularity that is used common for both PMI and CQI. For example, the signaling is via one parameter, for example, parameter formatIndicator indicating a value from {WBT, SBT}. In one example II.4.3.2, the F-T granularities of PMI and CQI are separate (can be the same or different), and the signaling/configuration is via 2 parameters (p1,p2), where p1=pmiFormatIndicator indicating a value from {WBT, SBT} for the F-T granularity of PMI, and p2=cqiFormatIndicator indicating a value from {WBT, SBT} for the F-T granularity of CQI. In one example II.4.3.3, the F-T granularities of PMI and CQI are separate (can be the same or different) and the signaling/configuration is via 1 parameter p indicating a pair (pmiFormatIndicator, cqiFormatIndicator).

(305) In one embodiment III.1, the granularity of RI reporting (included in the CSI reporting based on the codebook with Doppler component as described in this disclosure) is according to at least one of the following examples.

(306) In one example III.1.1, RI is not reported, and the CSI report (including components such as PMI and CQI) corresponds to a rank value, which can be fixed (e.g., rank 1), or configured (e.g., via higher layer signaling such as RI-restriction or rank-restriction).

(307) In one example III.1.2, a single RI value is reported, and the rest of the CSI components (such as PMI and CQI) are conditioned on (or according to) the rank value indicated via the reported RI value. The set of allowed or candidate rank values for RI reporting can be fixed (e.g., rank 1-2 or 1-4), or configured (e.g., via higher layer signaling such as RI-restriction or rank-restriction).

(308) In one example III.1.3, multiple RI values are reported, and for each rank value (indicated via one of the multiple RI values), the rest of the CSI

components (such as PMI and CQI) are reported, i.e., the CSI report includes $\{(r, C_{\text{sub}}.r), r \in R\}$, where R is the set of rank values indicated via the multiple RI values, r is a rank value included in the set R, and $C_{\text{sub}}.r$ is CSI components (such as PMI and CQI) conditioned on rank r. The set R can be determined based on a set S, which is the set of allowed or candidate rank values for RI reporting, and it can be fixed (e.g., rank 1-2 or 1-4), or configured (e.g., via higher layer signaling such as RI-restriction or rank-restriction). In one example III.1.3.1, the multiple RI values correspond to rank values potentially varying in time domain (e.g., across STs) and fixed in frequency domain (e.g., across SBs). Let $n_{\text{sub}}.t$ is the number of rank values reported via the multiple RI values. Then, the B or C or B+C time instances can be partitioned into $n_{\text{sub}}.t$ time-domain (TD) parts, and one rank value is reported for each TD part. Note that the rank value is fixed within each TD part (i.e., for all SBs and STs within each TD part). In one example III.1.3.2, the multiple RI values correspond to rank values potentially varying in frequency domain (e.g., across SBs) and fixed in time domain (e.g., across STs). Let $n_{\text{sub}}.f$ is the number of rank values reported via the multiple RI values. Then, the A frequency instances can be partitioned into $n_{\text{sub}}.f$ frequency-domain (FD) parts, and one rank value is reported for each FD part. Note that the rank value is fixed within each FD part (i.e., for all STs and SBs within each FD part). In one example III.1.3.3, the multiple RI values correspond to rank values potentially varying in both frequency domain (e.g., across SBs) and time domain (e.g., across STs). Then, the A frequency instances, and B or C or B+C time instances can be partitioned into $n_{\text{sub}}.t.n_{\text{sub}}.f$ TD-FD parts, and one rank value is reported for each TD-FD part. Note that the rank value is fixed within each TD-FD part (i.e., for all SBs and STs within each TD-FD part).

(309) In one example, when multiple RI values are reported, the RI reporting can be differential with respect to a reference RI value. As reference, one RI value is reported. As differential, one RI value is reported for each part, where a part is as explained in example III.1.3. For each part index p, a differential RI is defined as: $\text{RI offset level}(p) = \text{RI index}(p) - \text{reference RI index}$. An example of the mapping from the 2-bit differential RI values to the RI offset level is shown in Table 2.

(310) In one embodiment III.2, the granularity of RI reporting can be fixed (e.g., to example III.1.1 or III.1.2). Or it is signaled/configured (e.g., RRC, MAC-CE, or DCI). The configuration can include an information on whether RI reporting is according to example III.1.1 or example III.1.2 or example III.1.3. When the configuration allows RI reporting according to multiple of examples III.1.1 through example III.1.3, then the reporting format (or granularity) of RI reporting can be further configured, e.g., via a parameter riFormatIndicator . In one example III.2.1, the granularity of RI reporting is signaled/configured to be one of example III.1.a and III.1.b, where $(a,b) = \text{one of } (1,2), (1,3), (2,3)$. In one example III.2.2, the granularity of RI reporting is signaled/configured to be one of example III.1.1, example III.1.2, and example III.1.3. In one example III.2.3, the granularity of RI reporting is signaled/configured according to example III.2.1 or III.2.2 subject to UE capability reported by the UE. For example, whether the UE supports multiple RI reporting (example III.1.3) can be reported by the UE in its capability reporting, and only when the UE reports being capable of reporting multiple RI values, it can be configured with such RI reporting (by the NW.gNB).

(311) FIG. 18 illustrates a timing relation between CSI-RS occasion, CSI reference resource, and CSI report 1800 according to embodiments of the present disclosure. The embodiment of the timing relation between CSI-RS occasion, CSI reference resource, and CSI report 1800 is for illustration only. FIG. 18 does not limit the scope of this disclosure to any particular implementation of the timing relation between CSI-RS occasion, CSI reference resource, and CSI report 1800.

(312) The block error (BLER) probability requirement for CQI reporting as defined in Section 5.2.2.1 of TS 38.214 is as follows. The UE shall derive for each CQI value reported in uplink slot on the highest CQI index which satisfies the following condition: A single PDSCH transport block with a combination of modulation scheme, target code rate and transport block size corresponding to the CQI index, and occupying a group of downlink physical resource blocks termed the CSI reference resource, could be received with a transport block error probability not exceeding: 0.1, if the higher layer parameter cqi-Table in CSI-ReportConfig configures 'table1' (corresponding to Table 5.2.2.1-2, TS 38.214), or 'table2' (corresponding to Table 5.2.2.1-3, TS 38.214), or if the higher layer parameter cqi-Table in CSI-ReportConfig configures 'table4-r17' (corresponding to Table 5.2.2.1-5, TS 38.214), or 0.00001, if the higher layer parameter cqi-Table in CSI-ReportConfig configures 'table3' (corresponding to Table 5.2.2.1-4, TS 38.214).

(313) The time slot of the CSI reference resource is $n_{\text{sub}}.\text{ref} < n$. The CQI value is derived based on CSI-RS measurement occasions/window/interval $[k, k+W_{\text{sub}}.\text{meas}-1]$ prior to the CSI reference resource. An illustration of timing relation between CSI-RS occasion, CSI reference resource, and CSI report is provided in FIG. 18.

(314) In one embodiment IV.1, when the UE configured to report a CSI report that includes CQI reporting based on (or conditioned on) the PMI that indicates TD/DD components of the channel (as described in this disclosure), and when $W_{\text{sub}}.\text{CSI} > 1$, i.e., the CSI reporting/validity window $[l, l+W_{\text{sub}}.\text{CSI}-1]$ comprises multiple time slots or TD/DD units, then the CQI reporting and the corresponding BLER requirements are according to at least one of the following examples.

(315) In one example IV.1.1, the UE is configured to report one CQI value for entire time span $W_{\text{sub}}.\text{CSI}$ (i.e., WT as described earlier), and the BLER requirement is the same (one value), e.g., as in legacy Rel.15 NR specification (e.g., 0.1 or 0.00001).

(316) In one example IV.1.1.1, the CQI is expected to meet the BLER requirement at a time slot s within the $W_{\text{sub}}.\text{SCI}$ time slots, where the BLER requirement is the same as Rel.15 NR specification (i.e., 0.1 or 0.00001 as described above). In one example, s is fixed. In one example, $s = n_{\text{sub}}.\text{ref}$ (same as Rel.15 NR specification). In one example, $s = l + W_{\text{sub}}.\text{CSI} - 1$. In one example, $s = n_{\text{sub}}.\text{ref}$ when the codebook doesn't include TD/DD components (e.g., Rel.15/16/17 NR codebooks), and $s = l + W_{\text{sub}}.\text{CSI} - 1$ for when the codebook includes TD/DD components (e.g., new codebook as described in this disclosure). In one example, s is configured, e.g., from $\{n_{\text{sub}}.\text{ref}, l + W_{\text{sub}}.\text{CSI} - 1\}$ (2 values) or from the whole CSI reporting/validity window ($W_{\text{sub}}.\text{CSI}$ values).

(317) In one example IV.1.1.2, the CQI is expected to meet the BLER requirement for all slots in $W_{\text{sub}}.\text{SCI}$. For instance, the reported CQI is expected to meet a maximum of 0.1 (or 0.00001) BLER value during the whole CSI reporting window.

(318) In one example IV.1.2, the UE is configured to report one CQI value for entire time span $W_{\text{sub}}.\text{SCI}$ (i.e., WT as described earlier), and the BLER requirement depends on the slot index in which the CQI is derived.

(319) In one example IV.1.2.1, the BLER requirement is 0.1 for the reference slot $n_{\text{sub}}.\text{ref}$, and the BLER requirement is $v > 0.1$ for a future slot (after $n_{\text{sub}}.\text{ref}$), i.e., $n_{\text{sub}}.f = l + W_{\text{sub}}.\text{CSI} - 1 > n_{\text{sub}}.\text{ref}$. In one example, v is fixed (e.g., 0.15 or 0.2). In one example, v is configured (e.g., via RRC).

(320) In one example IV.1.2.2, the BLER requirement is $u < 0.1$ for the reference slot $n_{\text{sub}}.\text{ref}$, and 0.1 for a future slot (after $n_{\text{sub}}.\text{ref}$), i.e., $n_{\text{sub}}.f = l + W_{\text{sub}}.\text{CSI} - 1 > n_{\text{sub}}.\text{ref}$. In one example, u is fixed (e.g., 0.05 or 0.01). In one example, u is configured (e.g., via RRC).

(321) In one example IV.1.3, the UE is configured to report multiple (Y) CQI values (or multiple CQI values per SB in case of SB CQI reporting) which are for entire time span $W_{\text{sub}}.\text{SCI}$ (i.e., ST as described earlier), and the BLER requirement is the same (one value), e.g., as in legacy Rel.15 NR specification (e.g., 0.1 or 0.00001). In one example, Y is fixed. In one example, $Y = 2$. In one example, Y is determined based on TD/DD unit size (similar to SB size) or/and the value of B or C or B+C as described above. In one example, Y is configured (e.g., via RRC). The CSI reporting/validity window can be partitioned into Y parts (referred to as ST above, similar to SB), and one CQI value (or one CQI value per SB in case of SB CQI reporting) is determined for each part. Let $W_{\text{sub}}.\text{SCI},y$ be the number of time slots associated with the y-th part.

(322) In one example IV.1.3.1, for y-th part, the CQI is expected to meet the BLER requirement at a time slot $s_{\text{sub}}.y$ within the $W_{\text{sub}}.\text{SCI},y$ time slots associated with the y-th part $[l_{\text{sub}}.y, l_{\text{sub}}.y + W_{\text{sub}}.\text{SCI},y - 1]$, where the BLER requirement is the same as Rel.15 NR specification (i.e. 0.1 or 0.00001 as described above). In one example, $s_{\text{sub}}.y$ is fixed. In one example, $s_{\text{sub}}.y = l_{\text{sub}}.y$ (same as Rel.15 NR specification). In one example, $s_{\text{sub}}.y = l_{\text{sub}}.y + W_{\text{sub}}.\text{SCI},y - 1$. In one example, $s_{\text{sub}}.y$ is configured, e.g., from $\{l_{\text{sub}}.y, l_{\text{sub}}.y + W_{\text{sub}}.\text{SCI},y - 1\}$ (2 values) or from the whole CSI reporting/validity window for y-th part ($W_{\text{sub}}.\text{SCI},y$ values).

(323) In example IV.1.3.2, for y-th part, the CQI is expected to meet the BLER requirement for all slots in $W_{\text{sub}}.\text{SCI},y$. For instance, the reported CQI is expected to meet a maximum of 0.1 (or 0.00001) BLER value during the whole CSI reporting window for y-th part.

(324) In one example IV.1.4, the UE is configured to report multiple (Y) CQI values (or multiple CQI values per SB in case of SB CQI reporting) which are for entire time span $W_{\text{sub.CSI}}$ (i.e., ST as described earlier), and the BLER requirement depends on the slot index in which the CQI is derived. In one example, Y is fixed. In one example, $Y=2$. In one example, Y is determined based on TD/DD unit size (similar to SB size) or/and the value of B or C or $B+C$ as described above. In one example, Y is configured (e.g., via RRC). The CSI reporting/validity window can be partitioned into Y parts (referred to as ST above, similar to SB), and one CQI value (or one CQI value per SB in case of SB CQI reporting) is determined for each part. Let $W_{\text{sub.CSI},y}$ be the number of time slots associated with the y-th part.

(325) In one example IV.1.4.1, for y-th part, the BLER requirement is 0.1 for the slot $l_{\text{sub},y}$, and the BLER requirement is $v > 0.1$ for a future slot, i.e., $l_{\text{sub},y} + W_{\text{sub.CSI},y} - 1$. In one example, v is fixed (e.g., 0.15 or 0.2). In one example, v is configured (e.g., via RRC).

(326) In one example IV.1.4.2, the BLER requirement is $u < 0.1$ for the slot $l_{\text{sub},y}$, and 0.1 for a future i.e., $l_{\text{sub},y} + W_{\text{sub.CSI},y} - 1$. In one example, u is fixed (e.g., 0.05 or 0.01). In one example, u is configured (e.g., via RRC).

(327) Any of the above variation embodiments can be utilized independently or in combination with at least one other variation embodiment.

(328) FIG. 19 illustrates a flow chart of a method 1900 for operating a UE, as may be performed by a UE such as UE 116, according to embodiments of the present disclosure. The embodiment of the method 1900 illustrated in FIG. 19 is for illustration only. FIG. 19 does not limit the scope of this disclosure to any particular implementation.

(329) As illustrated in FIG. 19, the method 1900 begins at step 1902. In step 1902, the UE (e.g., 111-116 as illustrated in FIG. 1) receives a configuration about a CSI report, the configuration including information about a time interval W for the CSI report, the time interval W comprising multiple time slots; and a number (Y) of CQI values associated with the time interval W.

(330) In step 1904, the UE derives, based on the configuration, the Y CQI values, each of the Y CQI values at a corresponding time slot in the time interval W, wherein the Y CQI values satisfy a BLER probability requirement.

(331) In step 1906, the UE transmits the CSI report including the Y CQI values, wherein $Y \geq 1$.

(332) In one embodiment, the time interval W includes C time slots, $[l, l+C-1]$, no earlier than a time slot $n_{\text{sub.ref}}$ of a CSI reference resource, and the C time slots don't include a non-zero power CSI reference signal (NZP CSI-RS) resource for channel measurement.

(333) In one embodiment, the time interval W includes $B+C$ time slots, the B time slots, $[k, k+B-1]$, is no later than a time slot $n_{\text{sub.ref}}$ of a CSI reference resource, and includes at least one non-zero power CSI reference signal (NZP CSI-RS) resource for channel measurement, and the C time slots, $[l, l+C-1]$, is no earlier than the time slot $n_{\text{sub.ref}}$ of the CSI reference resource, and do not include a NZP CSI-RS resource for channel measurement.

(334) In one embodiment, $Y=1$ and a CQI value is derived at a time slot s in the time interval W, where $s > n_{\text{sub.ref}}$ and $n_{\text{sub.ref}}$ is the time slot of the CSI reference resource.

(335) In one embodiment, $Y > 1$, the time interval W comprises Y parts, $W_{\text{sub},1}, \dots, W_{\text{sub},y}$, and for each part $y \in \{1, \dots, Y\}$, one CQI value is derived at a time slot $s_{\text{sub},y}$ in the part $W_{\text{sub},y}$.

(336) In one embodiment, either $s_{\text{sub},y} > n_{\text{sub.ref}}$ for all y, or $s_{\text{sub},1} = n_{\text{sub.ref}}$ and $s_{\text{sub},y} > n_{\text{sub.ref}}$ for $y > 1$.

(337) In one embodiment, the configuration includes information about a CSI reference signal (CSI-RS) burst comprising $B \geq 1$ time instances of CSI-RS transmission, and the UE is configured to: measure the CSI-RS burst, and determine, based on the measurement of the CSI-RS burst, a precoding matrix indicator (PMI) and the Y CQI values, where the PMI indicates precoding matrices for time slots in the time interval W, and the Y CQI values are conditioned on the PMI, when the frequency domain (FD) granularity for the CQI is wideband, the Y CQI values are for a CSI reporting band, and when the FD granularity for the CQI is subband, the Y CQI values are determined for each subband (SB) in the CSI reporting band.

(338) FIG. 20 illustrates a flow chart of another method 2000, as may be performed by a base station (BS) such as BS 102, according to embodiments of the present disclosure. The embodiment of the method 2000 illustrated in FIG. 20 is for illustration only. FIG. 20 does not limit the scope of this disclosure to any particular implementation.

(339) As illustrated in FIG. 20, the method 2000 begins at step 2002. In step 2002, the BS (e.g., 101-103 as illustrated in FIG. 1), generates a configuration about a channel state information (CSI) report, the configuration including information about a time interval W for the CSI report, the time interval W comprising multiple time slots; and a number (Y) of channel quality indicator (CQI) values associated with the time interval W.

(340) In step 2004, the BS transmits the configuration.

(341) In step 2006, the BS receives the CSI report, wherein: the CSI report includes the Y CQI values, the Y CQI values are based on the configuration, each Y CQI value is at a corresponding time slot in the time interval W, the Y CQI values satisfy a block error (BLER) probability requirement, and $Y \geq 1$.

(342) In one embodiment, the time interval W includes C time slots, $[l, l+C-1]$, no earlier than a time slot $n_{\text{sub.ref}}$ of a CSI reference resource, and the C time slots don't include a non-zero power CSI reference signal (NZP CSI-RS) resource for channel measurement.

(343) In one embodiment, the time interval W includes $B+C$ time slots, the B time slots, $[k, k+B-1]$, is no later than a time slot $n_{\text{sub.ref}}$ of a CSI reference resource, and includes at least one non-zero power CSI reference signal (NZP CSI-RS) resource for channel measurement, and the C time slots, $[l, l+C-1]$, is no earlier than the time slot $n_{\text{sub.ref}}$ of the CSI reference resource, and do not include a NZP CSI-RS resource for channel measurement.

(344) In one embodiment, $Y=1$ and a CQI value is based on a time slot s in the time interval W, where $s > n_{\text{sub.ref}}$ and $n_{\text{sub.ref}}$ is the time slot of the CSI reference resource.

(345) In one embodiment, $Y > 1$, the time interval W comprises Y parts, $W_{\text{sub},1}, \dots, W_{\text{sub},Y}$, and for each part $y \in \{1, \dots, Y\}$, one CQI value is based on a time slot $s_{\text{sub},y}$ in the part $W_{\text{sub},y}$.

(346) In one embodiment, either $s_{\text{sub},y} > n_{\text{sub.ref}}$ for all y, or $s_{\text{sub},1} = n_{\text{sub.ref}}$ and $s_{\text{sub},y} > n_{\text{sub.ref}}$ for $y > 1$.

(347) The above flowcharts illustrate example methods that can be implemented in accordance with the principles of the present disclosure and various changes could be made to the methods illustrated in the flowcharts herein. For example, while shown as a series of steps, various steps in each figure could overlap, occur in parallel, occur in a different order, or occur multiple times. In another example, steps may be omitted or replaced by other steps.

(348) Although the present disclosure has been described with an exemplary embodiment, various changes and modifications may be suggested to one skilled in the art. It is intended that the present disclosure encompass such changes and modifications as fall within the scope of the appended claims. None of the description in this application should be read as implying that any particular element, step, or function is an essential element that must be included in the claims scope. The scope of patented subject matter is defined by the claims.

Claims

1. A user equipment (UE) comprising: a transceiver configured to: receive a configuration about a channel state information (CSI) report, the configuration including information about: a time interval W for the CSI report, the time interval W including C time slots, $[l, l+C-1]$, where l is a value of a slot index and C is a value of a time slot; and a number (Y) of channel quality indicator (CQI) values associated with the time interval W, where $Y \geq 1$; and a processor operably coupled to the transceiver, the processor configured to: derive, based on the configuration, the Y CQI values, each of the Y CQI values at a corresponding time slot in the time interval W, wherein: the Y CQI values satisfy a block error (BLER) probability requirement, and the transceiver is configured to transmit the CSI report including the Y CQI values, wherein the Y CQI values are reported in a same CSI report when $Y > 1$.

2. The UE of claim 1, wherein: the C time slots, $[l, l+C-1]$, are no earlier than a time slot $n_{\text{sub.ref}}$ of a CSI reference resource, and the C time slots do not include a non-zero power CSI reference signal (NZP CSI-RS) resource for channel measurement.
 3. The UE of claim 1, wherein $Y=1$ and a CQI value is associated with a time slot s in the time interval W .
 4. The UE of claim 1, wherein: $Y > 1$, the time interval W comprises Y parts, $W_{\text{sub}.1}, \dots, W_{\text{sub}.Y}$, and for each part $y \in \{1, \dots, Y\}$, one CQI value is associated with a time slot $s_{\text{sub}.y}$ in the part $W_{\text{sub}.y}$.
 5. The UE of claim 4, wherein: either $s_{\text{sub}.y} > n_{\text{sub.ref}}$ for all y , or $s_{\text{sub}.1} = n_{\text{sub.ref}}$ and $s_{\text{sub}.y} > n_{\text{sub.ref}}$ for $y > 1$, where $n_{\text{sub.ref}}$ is a time slot of a CSI reference resource.
 6. The UE of claim 1, wherein: the configuration includes information about a CSI reference signal (CSI-RS) burst comprising $B \geq 1$ time instances of CSI-RS transmission, and the processor is further configured to: measure the CSI-RS burst, and determine, based on the measurement of the CSI-RS burst, a pre-coding matrix indicator (PMI) and the Y CQI values, where the PMI indicates precoding matrices for time slots in the time interval W , and the Y CQI values are conditioned on the PMI, when the frequency domain (FD) granularity for the CQI is wideband, the Y CQI values are for a CSI reporting band, and when the FD granularity for the CQI is subband (SB), the Y CQI values are determined for each SB in the CSI reporting band.
 7. A base station (BS) comprising: a processor configured to generate a configuration about a channel state information (CSI) report, the configuration including information about: a time interval W for the CSI report, the time interval W including C time slots, $[l, l+C-1]$, where l is a value of a slot index and C is a value of a time slot; and a number (Y) of channel quality indicator (CQI) values associated with the time interval W , where $Y \geq 1$; and a transceiver operably coupled to the processor, the transceiver configured to: transmit the configuration; and receive the CSI report, wherein: the CSI report includes the Y CQI values, the Y CQI values are based on the configuration, each of the Y CQI values is at a corresponding time slot in the time interval W , the Y CQI values satisfy a block error (BLER) probability requirement, and the Y CQI values are reported in a same CSI report when $Y > 1$.
 8. The BS of claim 7, wherein: the C time slots, $[l, l+C-1]$, are no earlier than a time slot $n_{\text{sub.ref}}$ of a CSI reference resource, and the C time slots do not include a non-zero power CSI reference signal (NZP CSI-RS) resource for channel measurement.
 9. The BS of claim 7, wherein $Y=1$ and a CQI value is associated with a time slot s in the time interval W .
 10. The BS of claim 7, wherein: $Y > 1$, the time interval W comprises Y parts, $W_{\text{sub}.1}, \dots, W_{\text{sub}.Y}$, and for each part $y \in \{1, \dots, Y\}$, one CQI value is associated with a time slot $s_{\text{sub}.y}$ in the part $W_{\text{sub}.y}$.
 11. The BS of claim 10, wherein: either $s_{\text{sub}.y} > n_{\text{sub.ref}}$ for all y , or $s_{\text{sub}.1} = n_{\text{sub.ref}}$ and $s_{\text{sub}.y} > n_{\text{sub.ref}}$ for $y > 1$, where $n_{\text{sub.ref}}$ is a time slot of a CSI reference resource.
 12. A method for operating a user equipment (UE), the method comprising: receiving a configuration about a channel state information (CSI) report, the configuration including information about: a time interval W for the CSI report, the time interval W including C time slots, $[l, l+C-1]$, where l is a value of a slot index and C is a value of a time slot; and a number (Y) of channel quality indicator (CQI) values associated with the time interval W , where $Y \geq 1$; deriving, based on the configuration, Y CQI values, each of the Y CQI values at a corresponding time slot in the time interval W , wherein the Y CQI values satisfy a block error (BLER) probability requirement; and transmitting the CSI report including the Y CQI values, wherein the Y CQI values are reported in a same CSI report when $Y > 1$.
 13. The method of claim 12, wherein: the C time slots, $[l, l+C-1]$, are no earlier than a time slot $n_{\text{sub.ref}}$ of a CSI reference resource, and the C time slots do not include a non-zero power CSI reference signal (NZP CSI-RS) resource for channel measurement.
 14. The method of claim 12, wherein $Y=1$ and a CQI value is associated with a time slot s in the time interval W .
 15. The method of claim 12, wherein: $Y > 1$, the time interval W comprises Y parts, $W_{\text{sub}.1}, \dots, W_{\text{sub}.Y}$, and for each part $y \in \{1, \dots, Y\}$, one CQI value is associated with a time slot $s_{\text{sub}.y}$ in the part $W_{\text{sub}.y}$.
 16. The method of claim 15, wherein: either $s_{\text{sub}.y} > n_{\text{sub.ref}}$ for all y , or $s_{\text{sub}.1} = n_{\text{sub.ref}}$ and $s_{\text{sub}.y} > n_{\text{sub.ref}}$ for $y > 1$, where $n_{\text{sub.ref}}$ is a time slot of a CSI reference resource.
 17. The method of claim 12, wherein the configuration includes information about a CSI reference signal (CSI-RS) burst comprising $B \geq 1$ time instances of CSI-RS transmission, and the method further comprises: measuring the CSI-RS burst, and determining, based on the measurement of the CSI-RS burst, a pre-coding matrix indicator (PMI) and the Y CQI values, where the PMI indicates precoding matrices for time slots in the time interval W , and the Y CQI values are conditioned on the PMI, when the frequency domain (FD) granularity for the CQI is wideband, the Y CQI values are for a CSI reporting band, and when the FD granularity for the CQI is subband (SB), the Y CQI values are determined for each SB in the CSI reporting band.
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